Standard and Quality Practice

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1 Quality

Introduction

- This chapter aims to define quality as it applies to manufacturing and services industries, to introduce quality-related terminologies, and to establish a framework for the design and implementation of quality.
- Quality is a variable ideal that evolves with time. The improvement of quality is a never-ending process. Quality is a property or a characteristic.
- Quality is a relative phrase that typically implies to a product's final application. The significance of quality is its capacity to detect the specific needs of consumers, which aid in maintaining and expanding market share.

Quality

- There are a variety of definitions for the term 'quality', ranging from conventional to strategic. Conventional definitions of quality typically characterise a high-quality product as one that wears well, is well-made, and has a long life span. However, professionals competing in the global market are increasingly concerned with the strategic definition of quality that meets customer needs.
- Numerous authors have defined the concept of quality in a variety of ways. Garvin classifies the definition of quality into the following five categories: transcendent, product-based, user-based, manufacturing-based, and value-based.
- He also identifies an eight-attribute framework for defining quality—performance (basic operating characteristics of the product or service), features (characteristics that supplement basic functioning), reliability (consistency of performance and probability product working overtime), conformance (how well a product or service conforms to customer expectations and meeting pre-established standards), durability (expected product life), serviceability (speed, courtesy, competence, and ease of repair), aesthetics (appearance, feel, sound, smell, taste), and perceived quality (reputation and other indirect measures of quality).

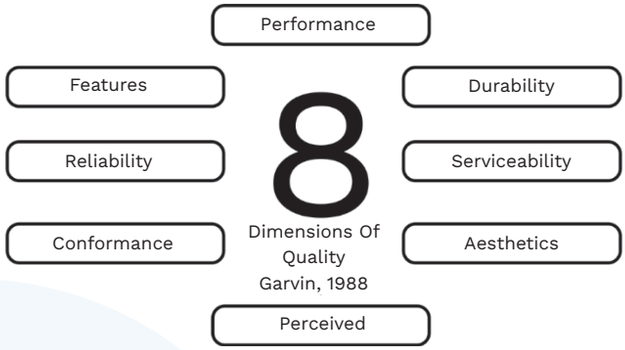


Fig. 1.1

- The frequently used definition for quality is attributed to Crosby (1979): "Quality is conformance to requirements or specifications." A more general definition proposed by Juran (1974) is as follows: "Quality is fitness for use." Hence, quality may be defined as follows: The quality of a product or service is the fitness of that product or service for meeting or exceeding its intended use as required by the customer. According to Edward Deming, "Quality is everyone's responsibility", and Aristotle, "Quality is not an act, it is a habit".



Fig. 1.2

Importance of Quality

- We cannot ignore how important it is for the business goals of a company to have quality control, quality assurance, and other quality documents set up and managed correctly.

They help the company get high-quality processes, procedures, systems, and people, which leads to high-quality products and services.

- Who decides how high a product or service's quality should be? Customers! Since customers' needs change, so should the quality. If, for example, customers want an air conditioner that has worked well for 15 years, that is exactly what a quality product should be. In this way, there is no one standard level of quality.
- The word 'quality' suggests that different groups of consumers have different levels of expectations. For example, some people think a good restaurant has amazing food and soft music playing in the background. But for another group of customers, a quality restaurant may have very different qualities: Buffets of delicious food with reasonable prices are available until early morning.
- Every manufacturing organisation is concerned with the quality of its product. While it is essential that quality requirements are satisfied and product schedules met, it is also equally important to consider the consequences of poor quality, such as Product failure, it goes over competitive pricing, and Loss of business: Poor designs or defective products, or poor services can result in loss of business.

Types of Quality

- To produce goods and services of consistent quality and cost, three types of quality are recognised. They are as follows:
  - Quality of design
  - Quality of conformance
  - Quality of performance
- **Quality of design** refers to the strict parameters that a product or service must possess, at minimum to meet customer requirements. It suggests that the product or service must meet the consumer's needs at a minimum level. For instance, a component with a drawing tolerance of ±0.002 mm would be seen as having a higher design quality than one with a tolerance of ± 0.02 mm.
- Product type, pricing, the profit policy of the company, product demand, the availability of parts and resources, and product safety all influence the design quality.
- **Quality of conformance** refers to the degree to which a manufactured product adheres to the quality of the design. Concerning the manufacturing industry, this phase focuses on the level of quality control from the procurement of raw materials to the distribution of final goods. It involves transforming user-based qualities into recognisable product properties.
- It comprises three major categories: defect prevention, defect detection, and defect analysis and correction. As its name implies, defect prevention is concerned with the means of preventing the incidence of defects. Statistical process control techniques are typically used to achieve defect prevention. Defects are identified by inspection, testing, and statistical analysis of process data. Lastly, the reasons for the presence of flaws are examined, and correction measures are adopted.
- **Quality of performance** refers to how well a manufactured object performs its intended function. It involves organising the production process to guarantee that product quality strictly corresponds to specifications. The objective of quality of performance studies is to discover how the quality characteristics identified in the design quality have been enhanced and innovated through the quality of conformance studies and market performance. Profitability is dependent upon market performance.

Cost of Quality

- The worth of a quality system is determined by its capacity to fulfil customer needs. In this sense, quality costs indicate the achievement or non-accomplishment of meeting product or service standards from the customer's perspective. This may include product design specifications, operating instructions, government rules, on-time delivery, marketing methods, and service commitments.
- The various components of quality costs are assigned according to product/service conformance or nonconformance. The attainment of requirements, identified as by-product or service conformance, has cost

component designated preventive costs, whereas nonconformance includes the cost components of appraisal and failure costs.

- In conclusion, quality costs can be interpreted as the difference between actual and reduced costs if all products and services conformed. This article discusses the four key cost categories for quality.

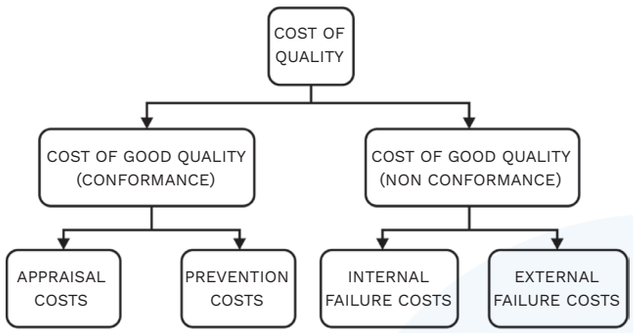


Fig. 1.3

Cost of Conformance

It is the cost of delivering items or services in accordance with the needed specifications. This might be termed productive use of time. Cost of conformance is further divided into two types

- Prevention costs
- Appraisal costs

Prevention Costs

- These are the costs associated with initiatives to prevent failure and efforts to eliminate flaws. Failure and appraisal costs are decreased as a result of preventive measures.
- Planning, implementing, and maintaining a quality system incurs expenditures to prevent poor product or service quality. Included are salary and development costs for product design, process and equipment design, process control techniques (such as control charts), information system design, and all other expenditures related to producing a product that meets specifications on the first try. In addition, costs linked with education and training are included. Additional costs include those connected with the origin and elimination of defects, process adjustments, and the price of a quality audit.
- The adoption of a quality system increases prevention costs, which may initially represent a major share of total quality costs. However, the pace of expansion slows over time. Even while preventative costs rise, they are more than balanced by reductions in total quality costs as a result of decreases in internal and external failure costs.

Appraisal Costs

- These are the costs associated with identifying potential defective products or services to ensure that no defects are discovered. Inspection costs, checking costs, auditing costs, surveying costs, prototype testing costs, vendor surveillance costs, etc., fall under the category of appraisal costs.
- Typically, appraisal costs occur during or after manufacturing, but before product delivery to the customer. Consequently, they are related to managing the outcome, whereas preventative costs are related to managing the intention or objective. Typically, appraisal costs decrease over time as more nonconformities are prevented.

Cost of Nonconformance or Failure Costs

- The cost of nonconformance or failure costs are associated with a process that does not operate according to the requirements. These costs occur due to defective parts or faulty services. It results from products or services not conforming to the requirements or customer or user needs.
- It can be divided into two types—internal failure costs and external failure costs.

Internal Failure Costs

- Internal failure costs are incurred when products, parts, materials, or services do not meet quality requirements before they are given to the customer. These costs would go away if the product or service had no problems.
- Internal failure costs include scrap and rework expenses for production-related materials, labour, and overhead. Correction of

nonconforming units, often known as rework, might incur additional production expenses, such as regrinding the outside diameter of an excessive component. If the outer diameter were too small, employing the component in the final product would be impractical and hence discarded.

- The cost of production time wasted owing to nonconformities must also be taken into account (e.g., if the poor quality of raw materials requires a retooling of equipment). In addition, depreciation costs, or the money lost because a faulty product must be sold at a lower price, are a component. As a whole quality management system is deployed and its effectiveness increases over time, internal failure costs will fall. As problems are prevented, less scrap and rework will result.

External Failure Costs

External failure costs are incurred if a product fails to work satisfactorily after ownership has been transferred to the client or if the services are substandard. This expense would be eliminated if no nonconforming units were manufactured. These costs include those incurred owing to customer complaints, including the costs of inquiry and modifications, as well as those related to receiving, handling, and replacing nonconforming products. Warranty charges (failure of a product within the warranty period) and product liability costs (costs or awards as a result of product liability litigation) are also included in this category. When a quality control system is successfully established, external failure costs are decreased.

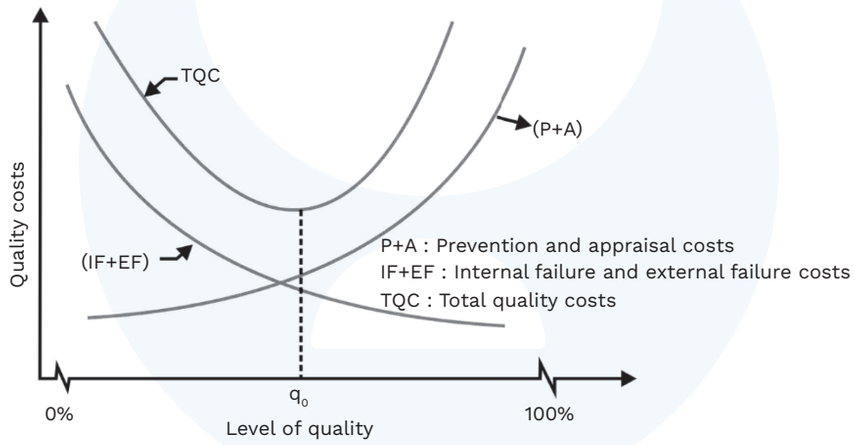


Fig. 1.4

Methods to Improve Quality

Brainstorming

- Brainstorming is a technique for generating ideas and exchanging knowledge to address a specific problem where participants are encouraged to explore without interruption.
- Brainstorming is a group exercise where participants share their ideas as they arise to them. Following the session, ideas are sorted and prioritised for subsequent action. It is a method for generating a free flow of problem-solving thoughts.

Quality Control Circles

- Quality control circles originated in Japan. It is a group of workers who voluntarily meet to discuss ways of improving products or processes. Quality circles are usually less structured and more informal than task forces entrusted with a project.
- The working staff involved in quality circles meet frequently at someone's home or the plant before the shift begins. The quality control circle team of people identifies problems, analyses data, recommends solutions and carries out management-approved changes.

- The basic concepts behind quality control circle activities, as part of company-wide quality control efforts, are inspecting quality at various points in the process, such as raw material and purchased parts, finished products, etc.

Benchmarking Process

- Benchmarking is applicable to practically every company and industrial activity. Improvements at all levels in certain areas will significantly contribute to market and financial success, whereas enhancements in other areas will have little impact. Organisations that benchmark modify the procedure to better suit their requirements and culture.
- The process of benchmarking includes identifying a critical process that needs improving, identifying an organisation that excels in this process, analysing the data of that organisation, and improving the critical process.
- Benchmarking includes identifying companies or other organisations that excel at something and analysing how they do it to learn how to enhance your own operation.

Quality Process

The quality process can be understood by three principles—quality control, quality assurance, and quality engineering.

Quality Control

- Quality control is the continual endeavour to preserve the integrity of a process to ensure the reliability of attaining an outcome.
- Quality control focuses on accomplishing quality requirements. As it relates to clinical trials, it comprises the operational processes and activities carried out inside the quality assurance system to verify that the quality requirements for trial-related activities have been met.
- Quality control is typically the responsibility of operational units, and quality is incorporated into outputs and validated as they are produced. Therefore, quality control is an inherent component of each operational unit's everyday operations.

Quality Assurance

- Quality Assurance is the planned or systematic action required to provide sufficient assurance that a product or service will meet the specified quality requirements.
- On the other side, quality assurance focuses on assuring that quality requirements are met. As it relates to clinical trials, it encompasses all planned and systemic measures created to guarantee that the trial is conducted and data are generated, documented (recorded), and reported in accordance with Good Clinical Practice (GCP) and the necessary regulatory standards.

Quality Control	Quality Assurance
Corrective tool	Managing tool
Product-oriented	Process-oriented
Reactive strategy	Proactive strategy
Detection of defects	Prevention of defects
Testing team responsibility	Everyone's responsibility
Performed after the final product is ready	Conducted in parallel with a project

Table 1.1

Quality Engineering

Quality engineering refers to the purposeful change of a process to improve the reliability of achieving an outcome.

Contribution of Quality Gurus

To properly comprehend the quality movement, it is necessary to examine the philosophies of influential persons who have influenced its evolution. Their qualitative and quantitative contributions were essential to the birth and growth of present understanding regarding quality. Although their ideologies differ, these

masters of quality assurance cite top-down leadership and devotion as a vital necessity for success. Their common goal is the constant improvement of each output, be it a product or a service, by eliminating undesirable variance and enhancing the underlying work processes. Their thoughts and teachings have contributed to our comprehension of quality. In the introduction, we have already covered some philosophies. Let us discuss some more.

S.No.	Quality Gurus	Their Contribution
1.	Edward Deming	Plan-Do-Check-Act (PDCA cycle) Fourteen points for transformation management
2.	JM Juran	Managerial practices, training, cost of quality
3.	Armand Feigenbaum	Total quality control
4.	Kaoru Ishikawa	Quality circles, seven quality control tools
5.	Philip Crosby	Zero defect, quality is free, cost of quality
6.	Genichi Taguchi	Quality loss function Robust design by 'Design of Experiments'

Table 1.2

Deming's Cycle

The continuous cycle of process improvement is centred on a scientific method that was once known as the Shewhart cycle, after its creator Walter A Shewhart. He, additionally, created control charts. The Japanese termed it the Deming cycle in the 1950s. It has four fundamental phases—plan, do, check, and act (the PDCA cycle).

- **Plan:** In this phase, improvement opportunities are identified and operationally specified. A framework addressing the effect of controllable process variables on process performance is established. Since customer satisfaction is the primary objective, the degree of disparity between customer needs satisfaction (as determined by market surveys and consumer research), and process performance (as determined by feedback information) is analysed. The objective is to diminish this disparity. The hypothesised links between the process variables and their effects on the outcome.
- **Do:** In this phase, the theory and strategy defined in the planning phase are implemented. Tests are performed in a laboratory or prototype environment. Customer and process-based feedback are received.
- **Check:** The results are now being examined. Does the proposed action narrow the gap between customer needs and process performance? Are there potential downsides associated with other customer-important quality characteristics? To obtain these answers, statistical approaches will be employed.
- **Act:** In the act phase, an implementation decision is taken. If the results of the analysis completed during the check phase are favourable, the suggested plan is accepted. After a full-scale implementation, customer and process input will be obtained again. This data will provide an accurate evaluation of the plan's success. If the findings of the check phase do not indicate considerable improvement, alternate plans must be created, and the cycle continues.
- Customers' needs vary throughout time. They alter due to time, competition, societal perspective, and other reasons. Therefore, the suggested plan is regularly evaluated to determine if it meets these requirements. This may necessitate additional modifications; hence, the cycle repeats, beginning anew with the planning phase.

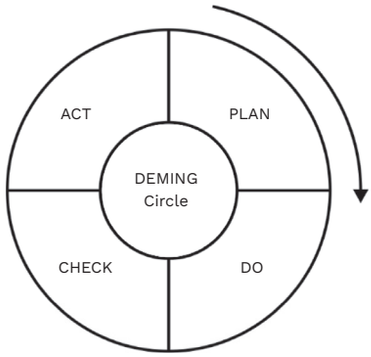


Fig. 1.5

PDSA Cycle (Plan–Do–Study–Act)

It is similar to Deming's PDCA and was created by Deming. By using the PDSA cycle in strategic planning, you can make sure that plans are made in a more organised way, that progress on plans is carefully tracked, that plans are changed when needed, that breakthrough goals are met, the planning process itself is standardised, that the planning process is constantly improved, and that organisational learning happens.

Practice Exercise

1. Which of the following would generally be considered as one of the costs of quality?
(A) Marketing costs
(B) Transaction costs
(C) Appraisal costs
(D) Overhead costs

2. The so-called 'quality gurus' of total quality management (TQM) do NOT include one of the following.
(A) Joseph M Juran
(B) Kaoru Ishikawa
(C) W Edwards Deming
(D) Bill Cosby

3. The concept of total quality control, i.e., that quality must be attended to at all stages of the industrial cycle and throughout the organisation, is the creation of which of the following pioneers?
(A) Genichi Taguchi
(B) Armand Feigenbaum
(C) W Edwards Deming
(D) Joseph M Juran

4. What is a quality circle's function?
(A) Bring management and employees together to find improvements.
(B) Managers and employees should be brought together to discuss output.
(C) Bring together management and employees to evaluate the quality of the work.
(D) Managers and employees should meet to discuss targets.

5. Two 'quality gurus' who were involved in training Japanese manufacturers in the 1950s were:
(A) Crosby and Shewhart
(B) Crosby and Juran
(C) Juran and Deming
(D) Deming and Crosby

6. Which of the following is a principle of total quality management?
(A) Continuous process improvement
(B) Continuous productivity improvement
(C) Continuous value enhancement
(D) Continuous defect elimination

7. The difference between an internal and an external customer is:
(A) Internal customers usually work in the business; external customers do not.
(B) External customers care about what they receive; internal customers do not.
(C) Internal customers do not evaluate quality, and external customers do.
(D) Quality-oriented businesses care only about external customers.

8. For services, a quality dimension that can be important would be:
(A) Deliverables
(B) Inputs
(C) Tangibles
(D) Durability

9. The most frequently used framework for obtaining continuous improvement is:
(A) Plan–do–check–act cycle
(B) Quality function deployment
(C) Quality circles
(D) Cause-and-effect process

10. Costs associated with efforts to prevent errors are:
(A) Appraisal costs
(B) Direct costs
(C) Prevention costs
(D) Expected costs

Answer Key

1. (C) **2.** (D) **3.** (B) **4.** (A) **5.** (C) **6.** (A) **7.** (A) **8.** (C) **9.** (A) **10.** (C)

2 Quality Control Tools

Introduction

- Karou Ishikawa, a Japanese expert, developed seven quality control tools to improve quality. He identified a set of tools that could be used to interpret data fully and derive maximum information from it. Ishikawa is also known as the father of quality control circles.
- Quality control tools are essential to the quality of the product and the improvement of service. These tools provide some very specific benefits, such as improved process information, improved communication, discussion based on facts, consensus for action, etc.

Quality Control Tools

The seven basic quality control tools used for problem-solving and process improvement are as follows.

- Histogram
- Flow chart
- Pareto diagram
- Check sheet
- Scatter diagram
- Cause and effect diagram
- Control chart

Histogram

A histogram is a graphical representation of data that subdivides a characteristic into classes or cells. It is similar to a bar chart. A histogram groups numbers into ranges. The range of each bar is also called the class interval. Quality control identifies the frequency of quality defect occurrence and displays quality performance. It is useful for finding out, any unusual values in the distribution and the range of those values. It organises data, evaluates process performance, and monitors process performance before and after a change.

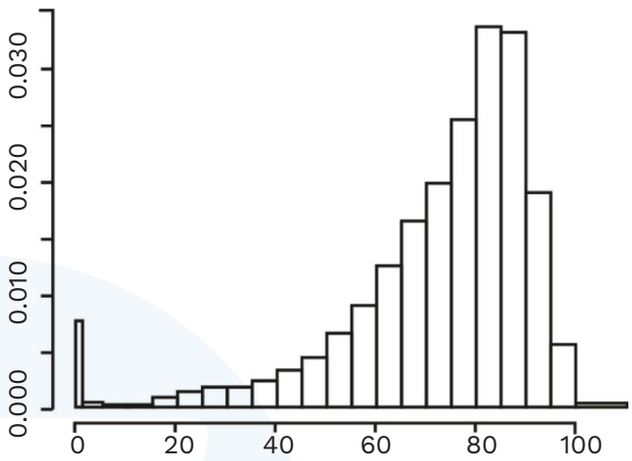


Fig. 2.1

Flow Chart

- A flow chart is a graphical representation of the stages and decisions required to complete a process. Each stage in the sequence is represented by a shape in the diagram, connecting lines, and directing arrows link the steps.
- It allows anyone to view the flow chart and logically follow the process from beginning to end. A flow chart is a powerful business tool with proper design and construction. It communicates the steps in a process very effectively and efficiently. It can be used to find quality problems.
- Flow Charts identify critical points in a process for data collection, locate bottlenecks, and redundant steps, improve problems, and control variations.

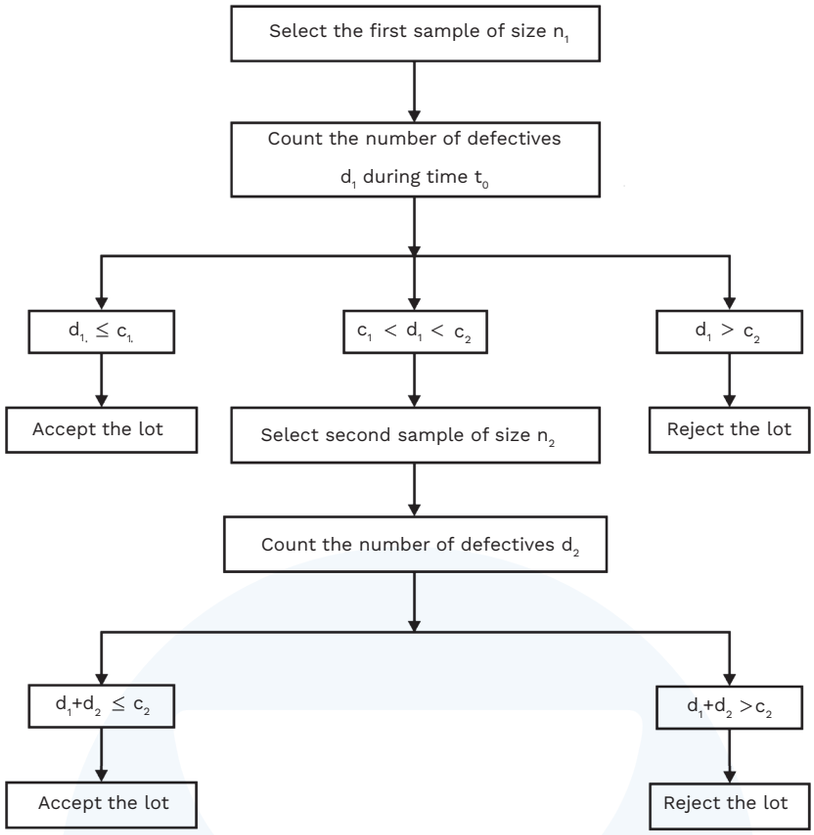


Fig. 2.2

Pareto Diagram

- In the process of quality improvement, Pareto diagrams are essential tools. Alfredo Pareto, an Italian economist, established that money is concentrated in the hands of a small number of individuals. This observation led to the development of the Pareto principle, which asserts that the bulk of wealth is held by a disproportionately small part of the population.
- In manufacturing or service businesses, for instance, problem areas or defect types have a comparable distribution. Among all the difficulties that occur, just a few are common; the rest are uncommon. These two problem areas are identified as the vital few and the trivial many.
- The Pareto principle also offers support to the 80/20 rule, which claims that 80% of issues (nonconformities or flaws) are caused by 20% of causes. By arranging problems in decreasing order of significance, Pareto diagrams help prioritise issues. In a resource-constrained context, these diagrams assist businesses in determining the order in which to tackle challenges.
- The diagram depicts a Pareto chart of airline customer displeasure causes. Forty percent of clients indicate that delays in arrival are the primary factor. Therefore, this is the issue that airlines must address first.

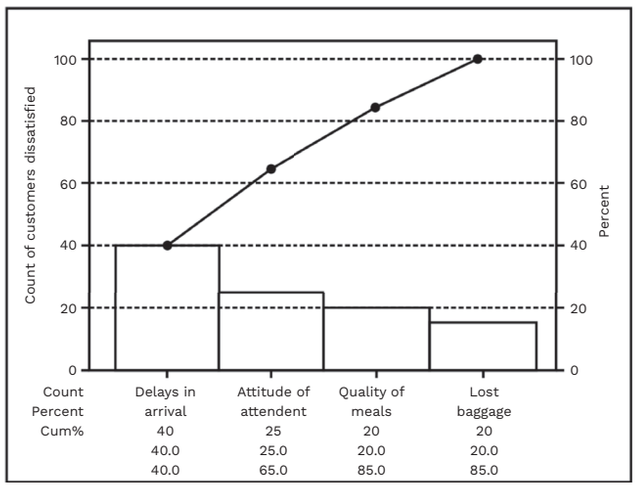


Fig. 2.3

Check Sheet

- It can be used to keep track of defects or make source people correctly collect data. Check sheets should not be compared with checklists because both are different.
- It is a data-gathering tool used for recording and organising data and is generally used in the preliminary stages of the study of a quality problem. It is useful for finding out any unusual values in the distribution and the range of those values.
- Check sheets can take many different forms, depending on the problem situation and the ingenuity of the analyst.

		Machine 1	Machine 2
Operator A	Morning	Y	Y
	Afternoon	YY	YYYYYY
Operator B	Morning	Y	YY
	Afternoon	YY	YYYYYYYYY

*Y = Number of times the engineer attended the machine in a day.

Table 2.1

Cause-and-Effect Diagram

- Cause-and-effect diagrams were created by Kaoru Ishikawa and are therefore commonly referred to as Ishikawa diagrams. Because of their look, they are also known as fishbone diagrams (in the plotted form).
- Cause-and-effect diagrams are used to identify and list methodically the various causes attributable to a problem (or an effect). These graphs help decide which of numerous causes has the most significant influence.
- A cause-and-effect diagram can help determine why a process goes out of control. Alternatively, if a process is steady, these diagrams might help management identify which causes to pursue process enhancement.
- The primary uses of cause-and-effect diagrams are cause enumeration, dispersion analysis, and process analysis.
- Cause enumeration is typically the result of a brainstorming session where all possible causes (no matter how remote) are identified to demonstrate their influence on the problem (or effect) at hand.
- In dispersion analysis, each major cause is investigated in depth by analysing the sub-causes and their impact on the quality feature (or effect) in question. This procedure is done in an ordered list for each main cause
- The cause-and-effect diagram helps us evaluate the reasons for any variability or dispersion. When constructing cause-and-effect diagrams for process analysis, the emphasis is placed on listing the causes in the order of operations execution. This procedure resembles the creation of a flowchart, with the exception that a cause-and-effect diagram specifies in detail the factors that influence the quality feature of interest at each stage of a process.

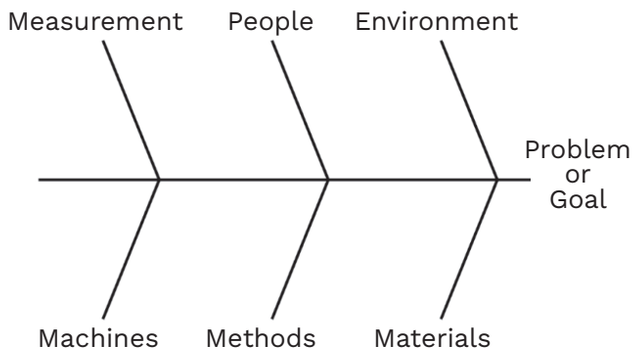


Fig. 2.4

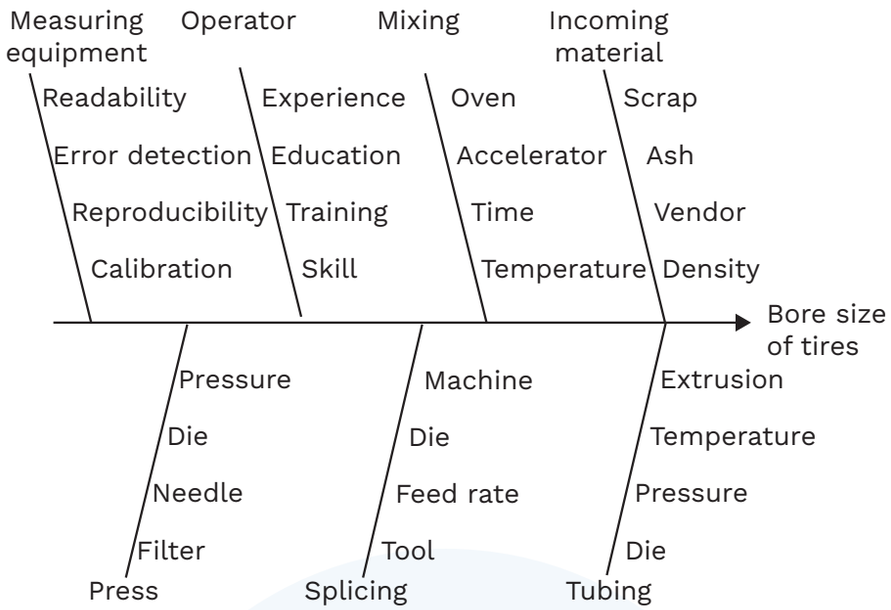
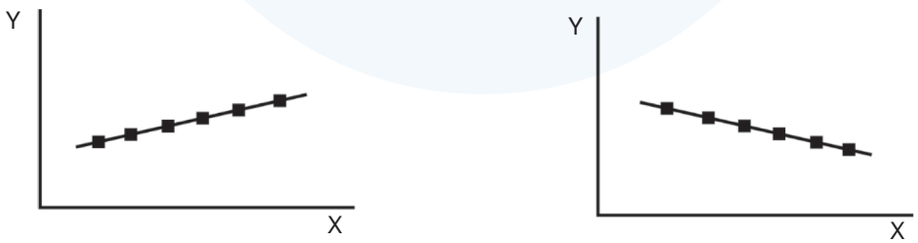


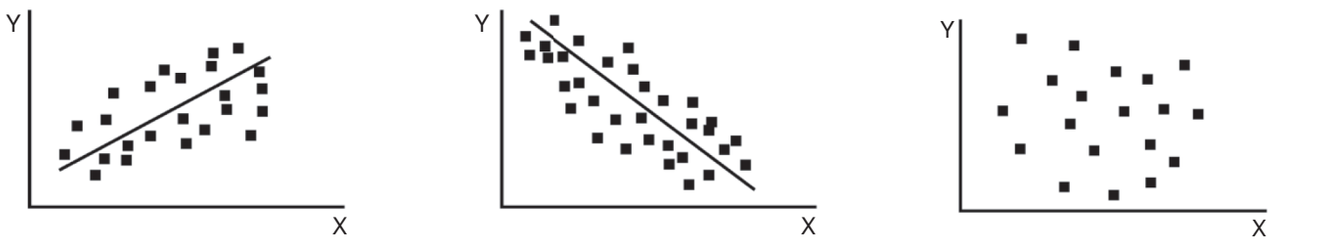
Fig. 2.5

Scatter Diagram

- The most basic form of a scatterplot is plotting bivariate data to illustrate the relationship between two variables. When analysing processes, the relationship between a controllable variable and the desired quality feature is often significant. This relationship may assist us in determining how to adjust a controllable variable to obtain a desired output characteristic level. As a follow-up to a cause-and-effect analysis, scatterplots are frequently employed.
- It is an x-y plot of data taken from the two variables. The data are plotted as pairs, i.e., for each x_i value, there is a corresponding y_i value. The shape of the data points considered often reveals a pattern or relationship between two variables.
- This relationship can fall into several categories: no correlation, curvilinear, positive correlation, and negative correlation.



(a) Perfect positive linear relationship, r=1 (b) Perfect negative linear relationship, r=-1



(c) Positive correlation of X and Y (d) Negative correlation of X and Y (e) No correlation between X and Y

Fig. 2.6

Control Charts

- It is a time-ordered values plot of sample statistics that assists in identifying assignable causes of variance in a process. It analyses the performance of a process over time, identifies variation in a process, and manages it.
- A control chart may indicate an out-of-control condition if one or more points exceed the control limits or if the plotted points display non-random patterns of behaviour. A control chart that has not triggered an out-of-control condition is considered stable and in a state of statistical control. Common-cause variation explains the variation displayed in the graph.

Variation

Variations can be divided into two parts.

- Common or random variation: These types of variation are always present, even under the best production conditions and minimal magnitude. They do not show a particular pattern and there is no problem regarding such variation.
- Assignable variation: These variations are of greater magnitude, and it denotes some problem in the production system due to which defective parts are produced. These may be due to machine setting, machine changing or tool wear, improper training of the operator, or defective raw material.

Types of Error

Errors can be divided into two types—type I error and type II error.

Type I Error

When there is no assignable cause, yet we think there is some problem, it is called a type-I error.

Type II Error

When we conclude that the universe has not changed when it has changed.

Statistical Process Control (SPC)

- A control chart is a graph that studies how a process changes over time and where data are plotted in time order.
- SPC is an online quality control tool. Control charts are used for process monitoring and variability reduction. Control charts can also be understood by the following figure.

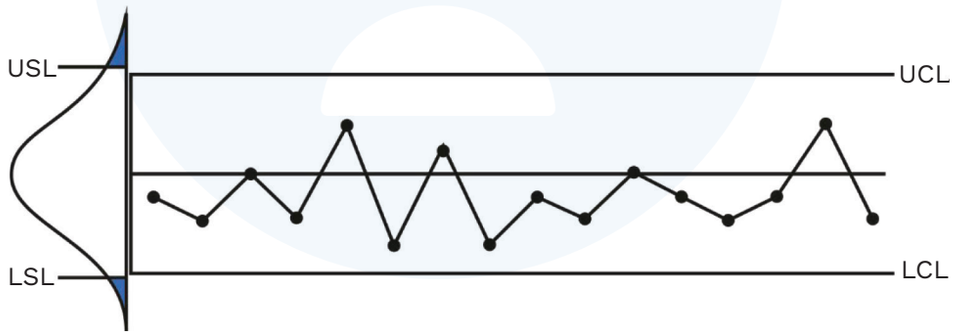


Fig. 2.7

- The above figure control chart has a central line for average, an upper line for upper control limit, and a lower line for lower control limit. Control chart variation can also be defined by the normal distribution curve.
- So, to decrease defective parts, we need to decrease the σ value. Normally ±3σ(6σ), control limits are selected most of the time for plotting control limits, and because of that, these charts are known as 3σ control charts.

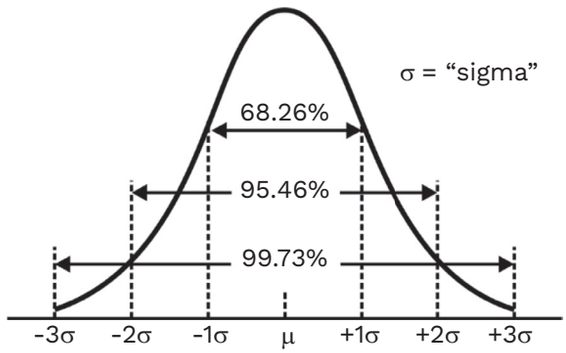


Fig. 2.8

Type of Control Charts

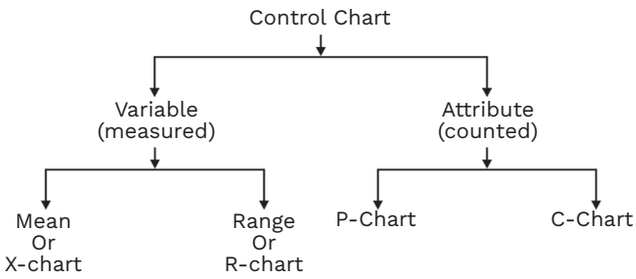


Fig. 2.9

Variable Control Chart

- These charts are applied to data that follow a continuous distribution and are measured on a continuous scale. These charts follow a normal distribution and can be in fractions or decimals like measurements of dimensions, weight, density, temperature, etc.
- Control charts for variables are often employed. Typically, they result in more effective control processes and provide more information on process performance than attribute control charts.

$\bar{x}$-Chart or Mean Chart

- This chart shows the centre of the process, in other words, it shows the variation in the average of samples. If we consider certain samples from a lot and measurement is done In those samples then, there will be some variations, and these variations can be better understood by the $\bar{x}$-chart.
- Construction of $\bar{x}$-chart: Several samples of components of a process are taken over a period of time and each sample consisting of several units is taken at random.

  Consider
  n = the Number of observations in each sample
  $\bar{x}$ = Mean of variation in each sample
  R = Range of variation of each sample
  Then for each sample,
  Mean $\quad \bar{x}_1 \quad \bar{x}_2 \quad \bar{x}_3 \quad \ldots \quad \bar{x}_n$
  Range $\quad R_1 \quad R_2 \quad R_3 \quad \ldots \quad R_n$
  Average of sample mean or grand average.

$$\bar{X} = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3 + \ldots + \bar{X}_N}{N} = \sum_{i=1}^{N} \frac{\bar{X}_i}{N}$$

From the values, we can calculate the central line, upper control limit (UCL), and lower control limit (LCL) of the $\bar{x}$ - chart

$CL = \bar{x}$

$UCL = \bar{x} + 3\sigma_{\bar{x}}$

$LCL = \bar{x} - 3\sigma_{\bar{x}}$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$\sigma_{\bar{x}}$ = standard deviation for a sample mean
σ = process or universe standard deviation,
Putting the value of $\sigma_{\bar{x}}$ in control limits,

$CL = \bar{x}$

$$UCL = \bar{x} + \frac{3\sigma}{\sqrt{n}}$$

$$LCL = \bar{x} - \frac{3\sigma}{\sqrt{n}}$$

R-Chart or Range Chart

- It shows the variation, in the range of samples, this chart is a measure of the spread of samples.
- Construction of R-chart: The ranges of each sample are calculated. The mean of the sample ranges is also calculated. The mean for the ranges gives the central line for the R chart.

$$\text{Average range, } \bar{R} = \frac{R_1 + R_2 + \ldots + R_n}{N} = \sum_{i=1}^{N} \frac{R_i}{N}$$

$CL = \bar{R} = \sigma d_2$

$UCL = \sigma d_2 + 3\sigma d_3$

$LCL = \sigma d_2 - 3\sigma d_3$

Now as $\bar{R} = \sigma d_2$

$$\sigma = \frac{\bar{R}}{d_2}$$

$$\text{Then, UCL} = \bar{R} + \frac{3\bar{R}d_3}{d_2} = \bar{R}\left(1 + \frac{3d_3}{d_2}\right) = D_4\bar{R}$$

$$LCL = \bar{R} - \frac{3\bar{R}d_3}{d_2} = \bar{R}\left(1 - \frac{3d_3}{d_2}\right) = D_3\bar{R}$$

$$D_4 = \left[1 + \frac{3d_3}{d_2}\right]$$

$$D_3 = \left[1 - \frac{3d_3}{d_2}\right]$$

and for $n < 7$, $D_3 = 0$

where D_3 and D_4 are constant factors and only depend on sample size n.

Attribute Control Chart

- These charts are counted and cannot be in fractions or decimals. These are applied when we want to find the presence or absence of something like success or failure, good or bad, defective or non-defective, etc. These data follow the binomial distribution and are applied for discrete data.
- A binomial distribution is a distribution whose parameters are n and p. All information about the binomial distribution can be calculated if the values of these parameters are known. This distribution applies to sampling without replacement from a population (or lot) that is larger than the sample size or sampling with replacement from a population that is finite. It is also used for scenarios where items are selected from an ongoing process (i.e., the population size is huge).

P-Chart (Proportion or Fraction Defective Chart)

- The P-larger than the sample size or sampling with replacement from a finite population chart is a control chart for attributes. It is intended to control the proportion or percentage of defectives per sample. The chart is most applicable when an article is inspected and classified as accepted or refused. These charts are used to monitor the proportion of items in a sample that is defective, and it is used for the situation when sample size n is not constant.

Average proportion defective,

$$\bar{p} = \frac{\bar{P}_1 + \bar{P}_2 + \ldots + \bar{P}_N}{N} = \sum_{i=1}^{N} \frac{\bar{P}_i}{N}$$

Average sample size,

$$\bar{\bar{n}} = \frac{\bar{n}_1 + \bar{n}_2 + \ldots + \bar{n}_N}{N} = \sum_{i=1}^{N} \frac{\bar{n}_i}{N}$$

$$CL = \bar{p}$$

$$UCL = \bar{p} + 3\sigma_p = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$$

$$LCL = \bar{p} - 3\sigma_p = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$$

np-Chart (Number of Defective Charts)

These charts are special cases of P-chart and are used for the condition when the sample size is constant, i.e., n = constant.

Then $n = n_1 = n_2 = \ldots n_N$

$CL = n\ \bar{p}$

$$UCL = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$$

$$LCL = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

c-Chart (Count of Defect Chart)

- It is used to monitor, the number of defects in a sample. It is used for the condition where we can only compute the number of defects but cannot compute the defective proportion.
- The c-chart technique helps to keep the defects per unit at the lowest limit. It follows Poisson's distribution
- A Poisson distribution describes the number of events that occur inside a given product unit, space or volume, or time interval. The assumption is that events occur independently and randomly.

According to Poisson's distribution,

Variance = Mean

Variance, $\sigma^2 = \bar{c}$

Standard deviation, $\sigma = \sqrt{\bar{c}}$

$CL = \bar{c}$

$UCL = \bar{c} + 3\sqrt{\bar{c}}$

$LCL = \bar{c} - 3\sqrt{\bar{c}}$

Solved Examples

1. What is the LCL for a p-chart when the average daily inspection quantity is 125 and the historical percentage of defectives is 0.10?

Solution:

$\bar{n} = 125$

$\bar{p} = 0 \cdot 10$

$(1-\bar{p}) = 1 - 0.10 = 0 \cdot 9$

For a p–chart

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$$

$CL = \bar{P}$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$$

$$LCL = 0 \cdot 1 - 3\sqrt{\frac{0.1 \times 0 \cdot 9}{125}}$$

$$= 0.1 - \frac{3 \times 0 \cdot 3}{5\sqrt{5}}$$

$$= 0.195 \cong 0.02$$

2. You manufacture a widget and use a c-chart to monitor the number of defects associated with the process. Your sample size is constant, and, on average, you find seven defects per sample. Identify the upper and lower control limit for the c-chart.

Solution:

$\bar{C} = 7$

In a C chart

$LCL = \bar{C} - 3\sqrt{C}$

$CL = \bar{C}$

$UCL = \bar{C} + 3\sqrt{C}$

$LCL = 7 - 3\sqrt{7} = 7 - 7.93 = -0.93$

$UCL = 7 + 3\sqrt{7} = 7 + 7.93 = 14.93$

3. Consider the following np-chart. What conclusion can be drawn?

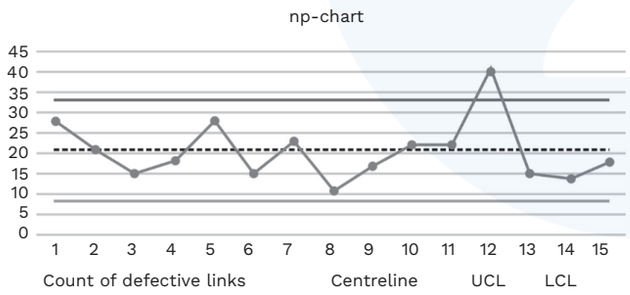


Fig. 2.10

Solution: In control

4. Which probability distribution is used to construct the a c-chart?

Solution:

Poisson distribution

5. What is the UCL for a p-chart when the average daily inspection quantity is 50 and the historical percentage of defectives is 0.057?

Solution:

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}}$$

$$= 0 \cdot 05 + 3\sqrt{\frac{0.05 \times 0 \cdot 95}{50}}$$

$= 0.14$

6. What type of control chart would be used to monitor the number of defectives for a process with a constant sample size?

Solution:

np-chart

7. You are manufacturing a widget and using an X-bar and R-chart to control the critical feature of the product. Your normal process has the following attributes:
X-double bar is 225, R-bar is 12, and n = 8.
Identify the upper and lower control limit for the range chart.

Solution:

$\bar{R} = 12$

If, $n \geq 7$ D_3 value exist

$N = 8$, $D_3 = 0.136$

$D_4 = 1.864$

For a $\bar{R}$ chart

$LCL = D_3 \bar{R}$

$CL = \bar{R}$

$UCL = D_4 \bar{R}$

$LCL = 0.136 \times 12 = 1.632$

$UCL = 1.864 \times 12 = 22.368$

8. Each month, the local finance department receives the same 20 reports. When a report has an error, it is sent back for correction. The finance department wants to track the number of reports initially denied each month. Which of the following control charts is most suitable?

Solution:

np-chart

9. You are constructing an NP chart, where you have sampled from 25 subgroups, each with 100 samples, and found a total of 145 defective units. Calculate the UCL for this process.

Solution:

$n = 100$

$\bar{p} = \frac{145}{25 \times 100} = 0.058$

$(1 - \bar{P}) = 1 - 0.058 = 0.942$

In a np chart

$CL = n\ \bar{p}$

$UCL = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$

$LCL = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$

$UCL = 100 \times 0.058 + 3\sqrt{100 \times 0.058 \times 0.942}$

$= 5.8 + 3\sqrt{5.8 \times 0.942}$

$= 12.79$

Practice Exercise

1. P-chart is:
(A) Variable control chart
(B) Attribute control chart
(C) Based on normal distribution
(D) Used continuous distribution

2. Which type of control charts are used for a variable?
(A) P–chart
(B) C–chart
(C) U–chart
(D) $\underline{X}$ –chart

3. Which of the following is incorrect for control charts?
(A) p and np control charts follow Poisson distribution.
(B) p and np control charts follow binomial distribution.
(C) c control chart follows Poisson distribution.
(D) None of these

4. Which of the following statement is incorrect?
(A) c-chart is based on Poisson distribution
(B) In Poisson distribution, mean and variance are equal.
(C) In Poisson distribution, the mean and variable are different.
(D) Reliability is the probability that an item will perform a required function without failure under stated conditions for a stated period of time.

5. Match List I (trend and defect) with List II (chart) and select the correct answer using the given codes.

List I	List II
A. Trend	I. R–chart
B. Dispersion	II. C–chart
C. Number of defects	III. $\underline{X}$ –chart
D. Number of defectives	IV. np–chart

Codes:

	A	B	C	D
(A)	III	I	II	IV
(B)	III	II	IV	I
(C)	IV	III	II	I
(D)	IV	II	I	III

6. Costs associated with efforts to prevent errors are:
(A) Preventive costs
(B) Expected costs
(C) Appraisal costs
(D) Direct costs

7. The most frequently used framework for obtaining continuous improvement is:
(A) Plan–do–check–act cycle
(B) Quality function deployment
(C) Quality circles
(D) Cause-and-effect process

8. For services, a quality dimension that can be important is:
(A) Tangibles
(B) Durability
(C) Inputs
(D) Deliverables

Answer Key

1. (B) **2.** (D) **3.** (A) **4.** (C) **5.** (D) **6.** (A) **7.** (A) **8.** (A)

3 Total Quality Management

Introduction

- Total quality management (TQM) is centred on three core themes: the client, the process, and the people. It is a disciplined approach to quality assurance within a business. Vision, mission, and management commitment are the pillars of TQM. They unite the consumer and the process in a unified whole.
- Customer satisfaction is the driving force of TQM; it is a significant concern. Without pleased clients, market share will not expand and revenues will not grow.
- Internal and external customers are a basic aspect of quality assurance initiatives. The customer of a processing unit is the department that receives a processed component. These internal customers' feedback indicates issue areas prior to the product's completion, hence lowering the cost of scrap and rework.
- The traditional approach or small 'q' is used as product-oriented and based on the post-production defect detection philosophy. It used to serve well under monopolistic situations where both the organisation and the market were treated as black boxes.
- Traditional quality management focuses on short-term earnings, stock price, lack of a defined strategic position in target sectors (poor competitive positioning), and squeezing costs, while tolerating high levels of waste, as well as, take-it-or-leave-it attitude towards consumers.
- However, the focus of the big 'Q' has a far broader scope and a strong customer orientation, which improves the product quality by enhancing the process.
- The big 'Q' describes the management of quality in all company operations and products. The small 'q' signifies management quality in a restricted capacity, historically solely incorporate products and procedures.
- The small 'q' approach to quality focuses on separating the best from the worst as opposed to eradicating flaws. The big 'Q' approach to quality emphasises the aspects of comprehensive quality management that are most closely related.

Improvement vs Control

- Control means maintaining a certain level of performance while improvement means achieving a higher level of performance. Generally, statistical process control (SPC) techniques are used to measure and analyse variations in a process.
- In any process, the measurable characteristic is subjected to random variations, due to common causes, around a certain target value, while due to some special causes, some non-random variations take place. Generally, control charts are used to identify these non-random deviations in a process caused by sporadic problems.

TQM Principles

TQM principles are a set of fundamental rules or beliefs for leading and operating an organisation. These are aimed at continually improving performances over the long-term, by focusing on customers and investors. It provides



Fig. 3.1

understanding and guidance on the application of quality management in an organisation. Following are the eight quality management principles that can facilitate the creation of a quality work.

Leadership

- Top management commitment is an important factor in the success of TQM programs. In an organisation, top management needs to set directions and create clear and visible quality values, high expectations, and a customer-oriented approach. These values, directions, and expectations need to address by all investors.
- Top management needs to ensure the creation of strategies, systems, and methods for achieving excellence strategies and values, that should help and guide all activities and decisions of the organisation.
- Top management must commit to the development of the entire workforce and should encourage participation, learning, innovation, and creativity by all employees.

Customer Satisfaction

- It refers to satisfying the expectations of customers. The customer is the driving force that provides feedback for error-free product development and production. The customer-focused organisations must produce products and services that are needed by the customer and provide satisfaction to the customers.
- This process can be achieved by actions, such asidentifying the customer's requirements and designing a product, that respond to customer's expectations, produce and deliver the product as per the design criteria, enhance after-sales service, etc.

Customer

- Anyone impacted by the product or service is called a customer. Several customers are waiting down the line when a product is being processed through several stages in an organisation before it reaches the final customer. Thus, we have people impacted within the organisation and also outside the organisation.
- There are two types of customers—internal and external.

Internal Customer

Any individual who receives and uses what a group or organisation provides. The individual inside a company who accepts the work of another and adds his or her own addition before passing it on to someone else.

External Customer

One who purchases a commodity or service. External customers are the driving forces behind an organisation's business.

Involvement of People

It is an approach to improve quality and productivity. The involvement of people at all levels enables full utilisation of their abilities for the organisation's benefit.

Process Improvement

Process improvement is the desired approach to achieve more efficient results when related sources and activities are managed as process mapping, analysing the process, and redesigning the process.

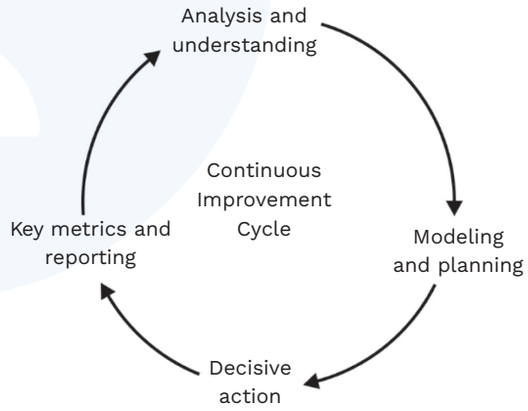


Fig. 3.2

Systems Approach Management

It is an approach to identifying, understanding and managing a system of interrelated processes for a given objective that contributes to the efficiency of the organisation.

Kaizen or Continuous Improvement

- It refers to the Kaizen's philosophy of striving for continuous improvement in all factors

related to the process of converting inputs to outputs.

- A permanent objective of an organisation is that it should continuously improve performance by addressing the needs of all interested parties.
- Targeting attainment of quality at the source. The philosophy of making each worker responsible for the quality of his or her work.

Steps in problem-solving based on Kaizen philosophy are:

- Define the problem and establish an improvement goal
- Collect data
- Analyse the problem
- Generate potential solutions
- Choose a solution
- Implement the solution
- Monitor the solution to see if it accomplished the goal

Factual Approach to Decision-Making

It is an approach that is based on the facts. Effective decisions are based on the logical or Intuitive analysis of data and information.

Mutually Beneficial Supplier Relationship

A hallmark of TQM is that it encourages long-term relationships with suppliers. Suppliers are considered partners in the whole process. The ability of the organisation and its suppliers to create value is enhanced by mutually beneficial relationships.

Taguchi's Approaches in TQM

Genichi Taguchi has an important influence on quality development, especially in the design area, both product, and process design. Taguchi has developed two methods for quality improvements.

- Robust Design
- Taguchi loss function

Robust Design

- This method helps to set specifications on product and process parameters to create a design that can resist failure or reduced performance in the face of variations.
- Taguchi says the variations can be noise factors. A noise factor is a source of variations that is impossible or difficult to control and affects the functional characteristic of the product.
- Noise factors can be described in three ways:
  - Unit-to-unit noise factors
  - Internal noise factors
  - External noise factors

Unit-to-Unit Noise Factors

These are random variations in every process and product caused by variability in materials, machinery, and workers. These are associated with a production process that is in statistical control.

Internal Noise Factors

These types of variations are internal to the product or process such as time-dependent factors, i.e., wear of mechanical components, spillage of raw materials, etc., and operational errors such as improper settings on the product or machine tool.

External Noise Factors

- These are the variations that are external to the product or process, i.e., outside temperature, humidity, raw material supply, and workers.
- Internal and external noise factors were previously known as assignable variations but divided both into parts, so one can easily understand the cause of variations.
- In process design, robustness means that the process continues to produce a good product with minimal effect from uncontrollable variations in its operating conditions.

Taguchi Loss Function

- Taguchi's loss function defines that "the loss of a product costs society from the time the product is released for shipment". It is a useful concept in tolerance design. Loss includes costs to operate, failure to function, maintenance and repair costs, etc.
- Defective products that are detected, repaired, reworked, or scrapped before shipment are not considered a part of this loss. Loss increases at an accelerating rate as the deviation grows.
- According to Taguchi, let, x = the quality characteristic of interest, T = its nominal

value, then the loss function will be a U-shaped curve as shown in the figure below.

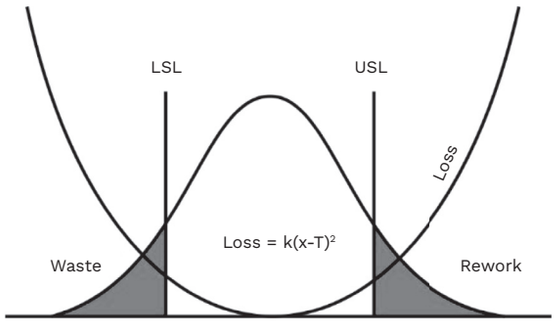


Fig. 3.3

- Taguchi uses a quadratic equation to describe the curve
  Loss function, $L(x) = k(x - T)^2$
  k = constant of proportionality
  loss ≠ 0 if within specifications
  = 1 if outside
  Mean square deviation, $MSD = \frac{\sum(x-T)^2}{n}$
- It is possible to make the calculation based on the Taguchi loss function if one accepts the assumption of the quadratic loss function.

Solved Examples

1. The quality loss function is given by $L(x) = k(x - N)^2$ what does 'k' stand for?
(A) Quality loss coefficient
(B) The quality characteristic of the selected product
(C) Tolerance
(D) Nominal value of the given product

Solution: (A)

2. Which among the following is the correct formula for evaluating the quality loss coefficient 'k'?
(A) k = c + d
(B) k = c/d
(C) $k = c/d^2$
(D) k = c – d

Solution: (C)

3. The quality loss function is given by $L(x) = k(x - N)^2$ what does 'N' stand for?
(A) Nominal value of the given product
(B) Tolerance
(C) Quality loss coefficient
(D) Quality characteristic of the selected product

Solution: (A)

4. The traditional approach focuses on specification limits and Taguchi's approach focuses on the target value.
(A) True
(B) False

Solution: (A)

5. Which parameter represents the slope of the quality loss function?
(A) Tolerance
(B) Quality loss coefficient
(C) Nominal value of the given product
(D) Quality characteristic of the selected product

Solution: (B)

6. Which among the following does not fit into the category of loss described by Taguchi?
(A) Maintenance and repair cost
(B) Customer dissatisfaction
(C) Failure to function
(D) Employee layoff

Solution: (D)

7. If tolerance of ±0.25 mm and the quality loss coefficient is 2520, then which of the following represented the quality loss function?
(A) $L = 2520 * (0.25)^2$
(B) L = 2520 * (0.25)
(C) L = 2520/(0.25)
(D) L = 2520 – (0.25)

Solution: (A)

8. If the specification of a product is 10 ± 3 for a particular quality characteristic and the average repair cost is Rs 200, what is the value of the quality loss coefficient?

(A) 20.23
(B) 22.23
(C) 24.23
(D) 26.23

Solution: (B)

$L(x) = 200$

$(x - T) = 3$

$L(x) = K (x - T)^2$

$K = \frac{200}{9} = 22.23$

9. Suppose the required length of your product is 1000 mm, and the customer will accept a tolerance of ±10 mm. Your loss is $10 if the part is rejected. You presently generate $3000 per month. Currently, they are roughly 1008 mm in length, which meets the requirements of the consumer.

You have a project that will cost $30000 and will shorten the product to a length of 1003 mm. As your product is currently being accepted by the client, management is hesitant to approve the project. Should you proceed?

Solution:

$L(x) = k (x - T)^2$ $x = 1008$

$= 0.1 (1008 - 1000)^2$ $T = 1000$

$= 6.4$ \$

For 300 units loss function will be

$3000 \times 6.4 = 19200$ \$

$L(x) = K (x - T)^2$ $x = 1003$

$= 0.1 (1003 - 1000)^2$ $T = 1000$

$= 0.9$ \$

For 300 units loss function will be

$3000 \times 0.9 = 2700$ \$

Saving in the 2nd case when compared to 1st case = \$19200 – \$2700 = \$16500

10. A machine produces 10,000 6'' long coil springs per day at a current production loss of \$30/part (overhead and rejection). Samples examined before and after improvement are as follows:

Length 6.1 5.8 6.3 6.4 5.7 (5 examples)

BEFORE:

Length 6.15 6.2 5.9 6.1 6.05 5.8 (6 examples)

AFTER:

Calculate expected savings.

Solution:

Length before

$MSD = \frac{\sum (x - T)^2}{n}, T = 6$

$$\frac{(6.1-6)^2 + (5.8-6)^2 + (6.3-6)^2 + (6.4-6)^2 + (5.7-6)^2}{5}$$

$= 0.0772$

$L(x) = K (MSD)$

$L(x) = 0.30$

$MSD = 0.772$

$K = \frac{0 \cdot 30}{0 \cdot 772} = 3.886$

Length after

MSD

$$= \frac{(6.15-6)^2 + (6.2-6)^2 + (5.9-6)^2 + (6.1-6)^2 + (6.05-6)^2 + (5.8-6)^2}{6} = 0.0204$$

$L(x) = K(MSD)$

$= 0.3886 \times 0.204$

$= 0.08$\$

Expected savings

$(0.3 - 0.08) \times 10000 = \$2200$

11. A part dimension is specified as 200 ± 0.4 mm, to investigate the impact of this tolerance on product performance, the company has studied its repair records to discover that if the ±0.4 mm tolerance is exceeded, there is a 50% chance that the product will be returned for repairs, at a cost of Rs 120. To the company (during the warranty period) or the customer (beyond the warranty period). What will be Taguchi loss function constant?

Old production status: loss = \$0.20 per part, MSD = 75

New production status: MSD = 0.50

Find \$ saving as a percentage of loss in the old production method.

Solution:

Old production

$L(x) = K (MSD)$

MSD = 0.75
L(x) = $0.20 per part
0.20 = K (0.75)
K = 0.266
New production
L(x) = K (MSD)
MSD = 0.50
= 0.266 (0.50) = 0.133
Saving as a percentage of loss old production method

$$\frac{0 \cdot 20 - 0.133}{0.20} \times 100 = 33.5\%$$

12. A part dimension is specified as 200 ±0.4 mm. To investigate the impact of this tolerance on product performance, the company has studied its repair records to discover that if the ±0.4 mm tolerance is exceeded, there is a 50% chance that the product will be returned for repairs, at a cost of Rs 120, to the company (during the warranty period) or the customer (beyond the warranty period). What will be Taguchi loss function constant?

Solution:
$L(x) = k (x - T)^2$
$(x - T) = 0.4$

$$\text{Loss function} = \frac{50}{100} \times \text{Rs } 120 + \frac{50}{100} \times \text{Rs } 0$$

L(x) = company customer
$L(x) = K(x - t)^2$
$60 = K (0.4)^2$
K = 375

Quality Function Deployment (QFD)

- Quality function deployment (QFD) is a tool for planning that focuses on developing quality into a product or service by incorporating client requirements. It is a systems approach incorporating cross-functional teams (whose members are not necessarily involved in product design) that examines the entire product development cycle. This quality cycle begins with the creation of a design that meets customer needs and continues through conducting detailed product analyses of parts and components to achieve the desired product, identifying the processes required to manufacture the product, developing product requirements, prototype testing, the final product or service testing, and troubleshooting after-sales.
- QFD is driven by client needs and turns them into acceptable technical requirements for products and services. It is aggressive. It is also known as the house of quality, matrix product planning, customer-driven engineering, and decision matrix, and it offers many benefits.
- It assesses competitors from two perspectives—the viewpoint of the customer and the technical perspective. The customer's perception of the company's competitors provides vital insight into the market potential of the company's products. The technical perspective, which is a sort of benchmarking, gives information on the relative performance of the company in comparison to market leaders. This study evaluates the extent to which products and processes require improvement and serves as a resource allocation guide.
- QFD shortens the product development cycle in all functional areas, from product conception and definition to production and sales. QFD reduces the time required for product redesign by taking into account product and design alongside manufacturing feasibility and resource constraints. Concerns regarding the product's process capability and after-introduction issues are reduced, as are midstream design modifications. This leads to major benefits for long-lead-time products. Consequently, QFD has proven crucial to the Ford Motor Company and General Motors' deployment of complete quality management.
- Companies utilise QFD to construct training programmes, select new personnel, establish supplier development criteria, and enhance service. QFD has also been utilised by cross-functional teams to illustrate the connections between departments, hence removing communication obstacles. Although the benefits of QFD are evident, its implementation necessitates a substantial

investment of time and manpower due to the vast quantity of data required.

- The phases of QFD include product planning, product design, process planning, and process control.

- There are many obstacles during the implementation of TQM. Some of them are listed below.

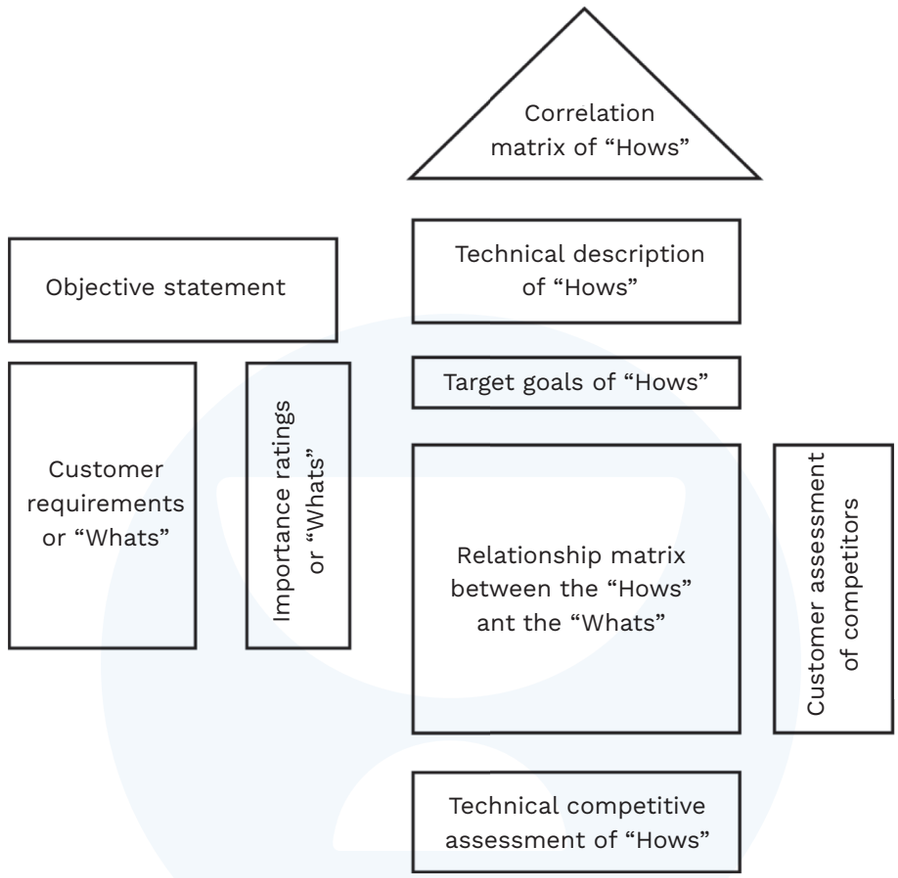


Fig. 3.4

- Lack of strategic planning related to changes necessary for various planning, processes, and products
- Poor customer focus
- Lack of communication among intra-departments of a company
- Lack of employee empowerment in the organisation
- More emphasis is given to financial results in the short-term
- Internal politics exist in the organisation
- Lack of strong motivation in employees
- Lack of commitment from top management

Malcolm Baldrige National Quality Award

Malcolm Baldrige was a former US Secretary of Commerce. This award is given for TQM. It helps to improve quality in US companies, recognises achievements of excellent firms, and provides examples to others.It establishes criteria for evaluating quality efforts.

Deming Prize

Deming prize is a highly coveted award in Japan. It was instituted in 1951 by the Union of Japanese Scientists and Engineers (JUSE) honouring W Edwards Deming.

Deming's 14 Points for TQM

- Constancy of purpose
- New philosophy
- Price tag awarding of business to suppliers
- Mass inspection
- Continuous improvement
- Training
- Supervision
- Fear
- Barriers
- Quotas
- Pride in workmanship
- Education
- Structure
- Slogans, etc.

Balanced Scorecard

Balanced scorecard (BSC) is a management method that incorporates strategy-derived measures. It combines measurements of both tangible and intangible assets. Through the creation of a communication and learning system, BSC focuses on achieving the company's objective. It translates the goal and strategy into objectives and measurements spanning four dimensions—learning and growth, internal processes, customers, and financial performance.

SERVQUAL

- The five dimensions of service quality refer to the SERVQUAL model of five important service dimensions, including reliability, assurance, tangibles, empathy, and responsiveness. The SERVQUAL is also known as the service quality model. It is a multidimensional research methodology designed to quantify the gap between customer expectations and perceptions of service quality across five dimensions.
- The SERVQUAL model refers to the five service quality characteristics that measure consumer expectations. It categorises the parts or components of service quality as the five essential service quality dimensions. Although the model's creators initially offered 10 characteristics of service quality, many experts ultimately settled on only five—reliability, assurance, tangibles, empathy, and responsiveness (RATER). After measuring the gaps, this model suggests the most typical reasons for service quality issues.

The Five Key Service Dimensions Of The Servequal Model

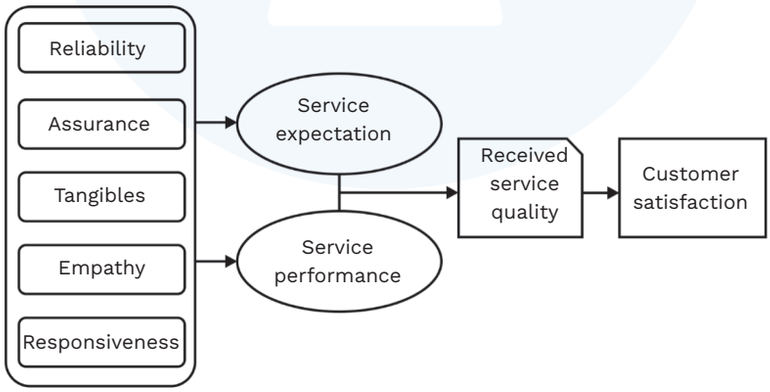


Fig. 3.5

4 Inventory Control

Inventory

Inventory is the stock of any item or resource used in an organisation. It is defined as stock in hand at a given point in time. It may be held for later use or to sell. Some examples of inventory are raw material, semi-finished goods, finished goods, etc.

Classification of Inventories

Inventories are generally classified into direct inventories, indirect inventories, seasonal inventories, lot size inventories, anticipation inventories, etc.

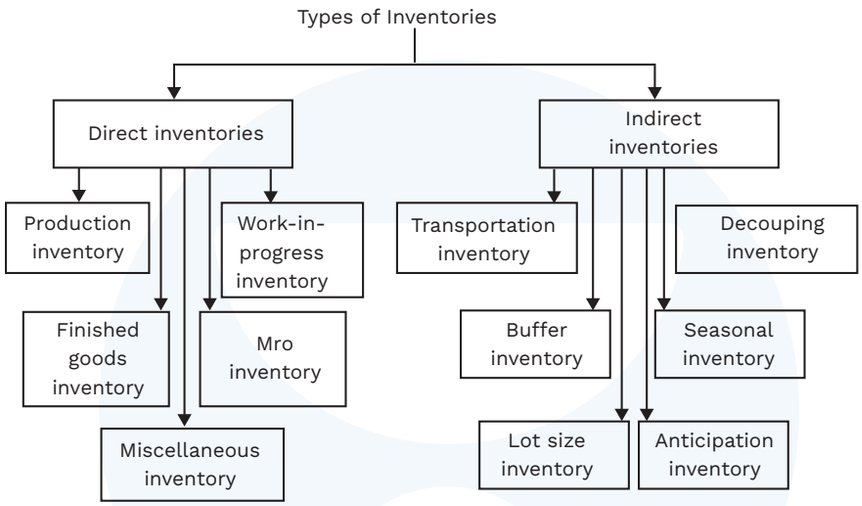


Fig. 4.1

Direct Inventories

Direct inventories are items that are directly used for production and are of different types, such as:

- Production inventory is items, such as raw materials and components that are used to produce the final product.
- Work-in-process inventory is items in semi-finished form or products at a different stage of production.
- Finished goods inventory includes the final products ready for dispatch to consumers or distributors.
- Maintenance, repair, and operating (MRO) inventory items are parts and consumable stores that do not go into the final product but are consumed during the production process.

Indirect Inventories

Indirect inventories are classified as follows.

- Transit or pipeline inventories, also known as movement inventories, consist of items that are currently under transportation
- Buffer inventories are required as protection against the uncertainties of supply and demand like supply chain failure, transportation delays, or unexpected surge in demand. Such situations require an extra stock of the item to reduce the number of stock-outs or back-orders. This extra stock over the average demand during the lead time is called buffer stock (or safety stock or cushion stock or contingency stock).
- Lot size inventories are usually purchased in lots to avail price discounts, reduce transportation and receiving costs, and

minimise handling and receiving costs. Lot size or cycle inventories are, therefore, held by purchasing items in lots rather than their exactly required quantities.

- Decoupling inventory is required to detach or disengage the various production systems components. For a product that must be processed on many different machines with varying processing periods, it is essential to have decoupling inventories of the product between the various units to maintain smooth and continuous production. In the event of fluctuating work rates, machine breakdowns or malfunctions, etc., the decoupling inventories operate as shock absorbers.
- Seasonal inventory items have seasonal demand, e.g., demand for ice cream and air conditioners in summer, raincoats in the rainy season, etc. Seasonal inventories must be maintained to meet the high seasonal demand.
- Anticipation inventory they is held to meet the anticipated demand.

Inventory Cost

Purchase Cost (PC)

It is the cost of purchasing an inventory item and it depends upon the quantity purchased.

PC = D × C

D = Annual demand
C = Cost/Unit (Rs/Unit)

Ordering or Setup Cost (OC)

- It is the cost associated with bringing inventory within the production system. It includes processing of purchase orders, transportation cost, quality inspection cost, etc.
- When the units are produced within the production system, then the cost associated with bringing shutdown production system, again into play is called setup cost (Cs). It includes maintenance costs, the cost associated with an arrangement of workers, raw materials, tools extra, and schedule chart preparation costs.

OC Number of ORDERS × Cost/Order

Holding or Carrying Cost (HC)

It is the cost associated with storage and keeping inventory items within the production system. It includes storage cost, handling cost, damage and depreciation cost, and Insurance cost. It depends upon the quantity and the period for which inventory is held.

HC = Average inventory × Cost of holding one unit

Shortage or Stockout Cost (SC)

Shortage simply means the absence of inventory and the cost associated with not serving the customer

SC = Number of units short × Shortage cost/Unit

Review System

Fixed Order System/Q System/Reorder Point System/Two Bin System

The re-order point is fixed and as the inventory level reaches that point fresh order for a fixed quantity is placed at that point. In this system, the size of the order is fixed while the time of the order is not fixed.

Periodic Review System/System/Fixed Period System

In this system inventory level is reviewed after a fixed period and fresh order is placed at that time. In this system time of order is fixed, while the size of the order is not fixed.

Types of Inventory Models

- In the deterministic model, demand and lead time (the amount of time between a purchase order to replenish products and the order received at the warehouse) are known with certainty and are constant. So, there is no need to carry safety or buffer stock.
- In the probabilistic model, demand and lead time are not known with certainty and are not constant. So, this model needs safety stock to prevent a stock out.

Deterministic Model

Harris–Wilson Model (EOQ Model)

Ford Harris developed this model in 1915. RH Wilson 1943 popularised this model among

researchers and practitioners. The following notations are used in this model.

D = Annual demand of inventory item (units/year)
Q = Quantity to be ordered at each order point (units/order)
d = Demand or consumption rate (units/time)
C = Unit cost of inventory (Rs/unit)
C_o = Cost of placing an order (Rs/order)
C_h = Cost of holding one unit in inventory for 1 year (Rs/unit/year)
LT = Lead time
T = Time

The assumptions made in the EOQ model are demand for inventory is known with certainty, all orders are placed in a single lot, no shortages or stock out are allowed, no quantity discounts are allowed, inventory is controlled from one point in the system, and lead time is assumed to be zero.

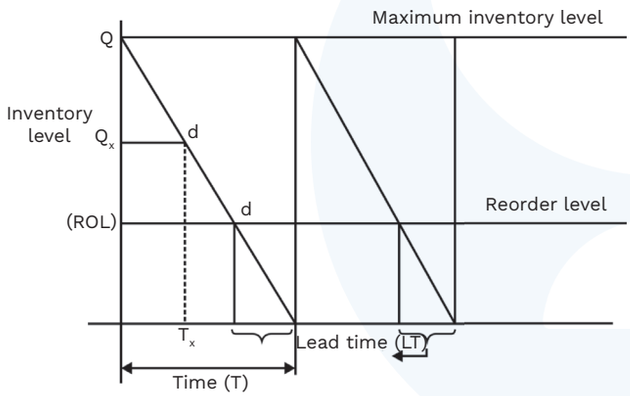


Fig. 4.2

$Q = T \times d$

Reorder level (ROL) = $LT \times d$

At any inventory value (x) $Q_x = T_x \times d$

$$\frac{Q}{T} = d = \frac{RoL}{LT} = \frac{Q_x}{T_x}$$

Total annual cost = Purchase cost + Ordering cost + Holding

Annual purchase cost = DC

Annual ordering cost (OC) = Number of orders in a year $\times C_o$

$$\text{Number of orders (N)} = \frac{D}{Q} = \frac{\text{Annual Demand}}{\text{Units ordered / Year}}$$

$$\text{Annual ordering cost} = \frac{D}{Q}.C_o$$

Average holding cost (HC) = Average inventory for the year $\times C_h$

$$\frac{Q}{2} \cdot C_h$$

Total inventory cost (TIC) = OC + HC

$$\frac{D}{Q} \cdot C_0 + \frac{Q}{2} C_h$$

Total annual cost (TAC) = TIC + DC

$$DC + \frac{D}{Q} \cdot c_0 + \frac{Q}{2} \cdot C_h$$

The economic ordering quantity (Q^* or EOQ) where holding cost (HC) becomes equal to ordering cost (OC).

OC = HC

$$\frac{D}{Q^*} \cdot C_0 = \frac{Q^*}{2} C_h$$

$$Q^* = \sqrt{\frac{2DC_0}{C_h}}$$

Total inventory cost at EOQ,

$$TIC^* = \frac{D}{Q^*} \cdot C_0 + \frac{Q^*}{2} C_h$$

$$\frac{D}{Q^*} \cdot C_0 = \frac{Q^*}{2} C_h$$

$$TIC = \sqrt{2DC_0C_h}$$

$$TIC = \frac{D}{Q^*} \cdot C_0 + \frac{Q^*}{2} C_h$$

The optimum number of orders per year

$$N^* = \frac{D}{Q^*}$$

The optimum length of inventory cycle or optimum time between two successive orders.

$$T^* = \frac{1}{N^*} = \frac{Q^*}{D}$$

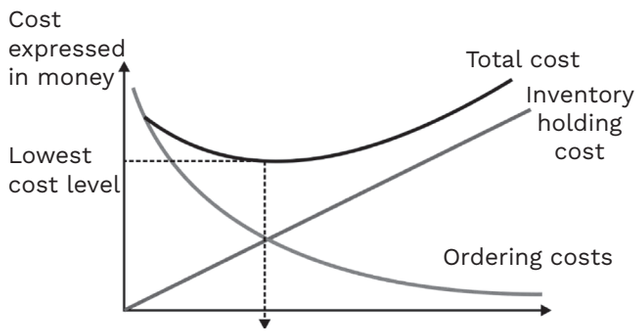


Fig. 4.3

Robustness is simply the ratio of total inventory cost at any point to the total inventory cost corresponding to EOQ. It indicates how sensitive our inventory model is for different sizes.

$$\text{Robustness} = \frac{TIC}{TIC^*}$$

EOQ with Price Break or Quantity Discount

- In some conditions, discounts are given on the unit purchase price of inventory due to large quantity purchases. These discounts take the form of a price break. A discount is always given on the unit purchase price of inventory in this model. The purchasing cost is also considered, along with ordering and holding costs, to determine the best order size.
- The procedure is first to find feasible EOQ and then the total cost is computed at feasible EOQ and the price breakpoint having an order size more than EOQ, wherever the total cost is the least, is the best order size.

Total cost = Purchase cost + Ordering cost + Holding cost

$$TC = DC + \frac{D}{Q}C_0 + \frac{Q}{2}C_h$$

Example

For 600 items

$= 100 \times 70 + 400 \times 600 + 100 \times 50$

P → 1 – 100 units → 70 Rs per unit

Q → 100 – 501 units → 60 Rs per unit

R → 501 – 1000 units → 50 Rs per unit

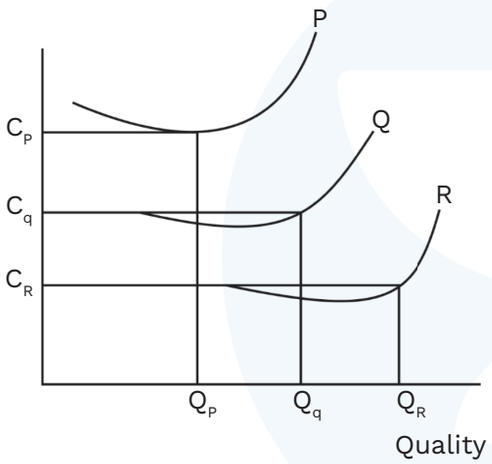


Fig. 4.4

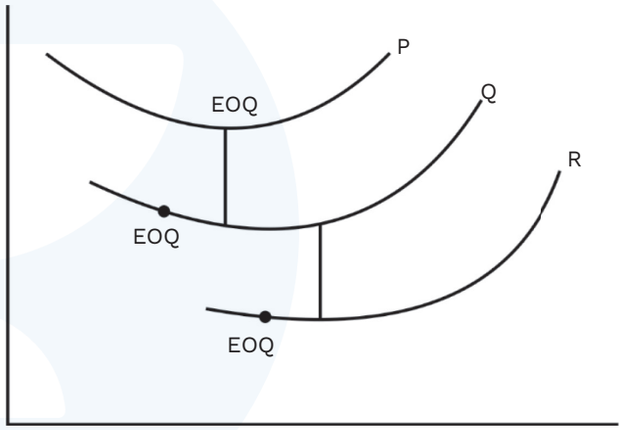


Fig. 4.5

Production–Consumption or Build-up Model

- If we make an inventory with production rate (p) and consumption with demand rate (d) then
- (p – d) inventory build-up, then production will be stopped once the maximum inventory is reached then only consumption happens. The following diagram explains the production-consumption model

$$Q = t_P \times P = T \times d$$

$$t_P = \frac{Q}{P}$$ (t_p is production or manufacturing time)

P = production rate, d = demand or consumption rate, and Q = quality produced

Maximum inventory $Q_m = t_p \cdot (p - d)$

$$Q_m = \frac{Q}{p}(p - d) = Q\left(\frac{p - d}{p}\right)$$

Total inventory cost = Ordering cost (setup cost) + Holding cost

TIC = OC + HC

HC = Average inventory x C_h

$$TIC = \frac{D}{Q}C_0 + \frac{Q}{2}\left(\frac{p - d}{P}\right)C_h$$

To find the best order size (EOQ)

$$\frac{d(TIC)}{dQ} = 0$$

$$\frac{d(TIC)}{dQ} = \frac{1}{2}\left(\frac{p - d}{P}\right) - \frac{D}{Q^{*2}}C_0 = 0$$

Economic order quality $Q^* = \sqrt{\frac{2DC_0}{c_h}\left(\frac{p}{P-d}\right)}$

At EOQ, OC = HC

Total inventory cost at EOQ

$TIC^* = \sqrt{2DC_0C_h} \cdot \sqrt{\frac{P}{P-d}}$

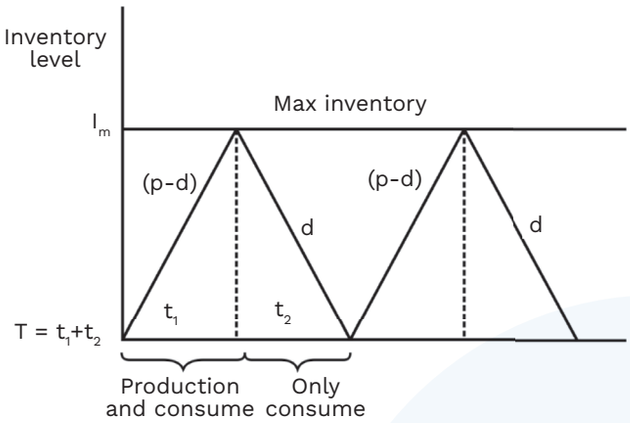


Fig. 4.6

Shortage or Back Order Model or Stock Out

- Back order model is similar to the basic EOQ model. The only difference is that here shortage is allowed.
- Planned shortage or back order is the situation in which a customer places an order and finds that inventory is out of stock, then the customer waits for the next shipment, to get their order fulfilled.
- Shortages also have certain advantages like shortages increase the cycle time and thus decrease the ordering cost. Shortages also result in decreasing the net stock in inventory thus reducing the holding cost.

  Total inventory cost TIC = OC + HC + SC

Order Cost $OC = \frac{D}{Q} \cdot C_0$

Holding Cost $HC = \frac{(Q-S)^2}{2Q} \cdot C_h$

Shortage cost $SC = \frac{S^2}{2Q} C_b$

C_b = back order or shortage cost per unit back-ordered per year (Rs/unit/year),

S = no. of units that were back-ordered.

Total inventory cost TIC

$= \frac{D}{Q} \cdot C_0 + \frac{(Q-S)^2}{2Q} \cdot C_h + \frac{S^2}{2Q} \cdot C_b$

Economic order quantity,

$Q^* = \sqrt{\frac{2DC_0}{c_h}\left(\frac{C_b + C_h}{C_b}\right)}$

Total inventory cost at EOQ, TIC^*

$= \sqrt{2DC_0C_h\left(\frac{C_b + C_h}{C_b}\right)}$

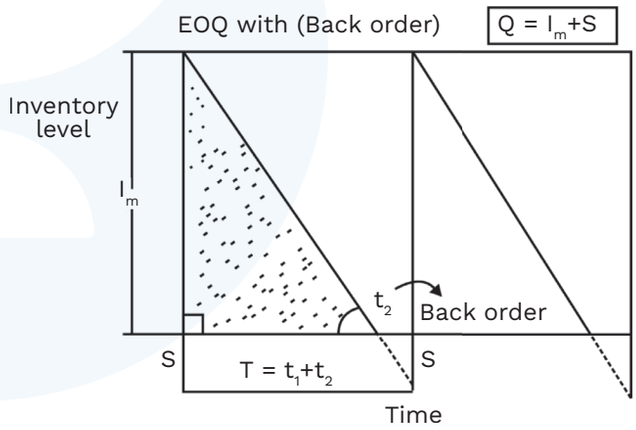


Fig. 4.7

Solved Examples

1. Market demand for springs is 800000 per annum. A company purchases these springs in lots and sells them. The cost of making a purchase order is Rs 1200. The cost of storage of springs is Rs 120 per stored piece per annum. What is the economic order quantity?

(A) 400

(B) 2828

(C) 4000

(D) 8000

Solution: (C)

D = 800000

C_o = Rs 1200 per order

C_H = Rs 120 per unit year

$Q^* = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 800000 \times 1200}{120}} = 4000$

2. If the annual demand = 18000 units, purchasing per unit = Rs 4, ordering cost per order = Rs 120, inventory carrying cost = 12% of purchasing price per unit/unit/yr, what will be the saving in the total cost per year if the orders are placed as per EOQ in place of four orders per year?

(A) Rs 120
(B) Rs 240
(C) Rs 1000
(D) Rs 360

Solution: (A)

D = 18000

C = Rs 4 per unit

C_o = Rs 120 per order

$C_H = \frac{12}{100} \times \text{Rs } 4$

$$Q^* = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 18000 \times 120}{\frac{12}{100} \times 4}} = 3000$$

$$N = \frac{D}{Q^*} = \frac{18000}{3000} = 6$$

$$N = 4,\ N = \frac{D}{Q} = 4 = \frac{18000}{Q} \qquad Q = 4500$$

(A) TAC = PC + OC + HC

$$N = 6, \quad = \frac{D}{Q}C_0 + \frac{Q}{2}C_H$$

EOQ = 3000

$$= 6 \times 120 + \frac{3000}{2} + \frac{12}{100} \times 4 = 1440$$

(B) TAC = PC + OC + HC

$$N = 4, \quad = \frac{D}{Q}C_0 + \frac{Q}{2}C_H$$

EOQ = 4500

$$= 4 \times 120 + \frac{4500}{2} + \frac{12}{100} \times 4 = 1560$$

Then, 1560 – 1440 = Rs 120

3. In the basic EOQ model, if the lead time increases from 5 to 10 days, the EOQ will be:

(A) Double
(B) Decreased by a factor of two
(C) Remains the same
(D) The data is insufficient to find EOQ

Solution: (C)

4. Consider the following statements about the inventory review system.

I. In the P system, the time between the placing of orders is not fixed, whereas in the Q system, the time between the placing of orders is fixed.
II. In the Q system, the quality to be ordered is not fixed, whereas in the P system, the quantity to be ordered is fixed.

Which of the above statements are correct?

(A) I only
(B) II only
(C) Both I and II
(D) None of the above

Solution: (D)

5. If orders are placed once a month to meet an annual demand of 6000 units, then what will be the average inventory?

(A) 200
(B) 250
(C) 300
(D) 500

Solution: (D)

Average inventory = Q/2 = 250

N = D/Q

12 6000/Q

Q = 500

6. In the basic EOQ model, if demand is 60 per month, the ordering cost is Rs 12 per order, the holding cost is Rs 10 per unit per month, then what is the EOQ?

(A) 12
(B) 24
(C) 144
(D) 28

Solution: (A)

D = 60 × 12

C_o = Rs 120 per order

C_H = 10 * Rs 12

$$EOQ = \sqrt{\frac{2 \times 60 \times 12 \times 12}{120}} = 12$$

7. Annual demand for a product costing Rs 100 per piece is 900. Ordering cost per order is Rs 100 and inventory holding cost is Rs 2 per unit year. The economic lost size is:

(A) 200
(B) 300

(C) 400
(D) 500

Solution: (B)

D = 900

C = 100

C_o = Rs 100 per order

C_H = Rs 2

$$EOQ = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 900 \times 100}{2}} = 300$$

8. An industrial machine tools manufacturer suppliers' replacements from its inventory for a particular part, the annual demand is expected to be 750 units. Machine setup costs are $50, carrying costs are 25% per year, and the part is valued in inventory at $35 each. The economic order quantity is _____ units.

Solution:

D = 750

C_o = 50

$$C_H = \frac{25}{100} \times 35$$

$$EOQ = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 750 \times 4 \times 50}{35}} = 93$$

9. A local tyre distributor expects to sell approximately 9600 steel belted radial tyres next year. Annual carrying cost is Rs 16 per tyre and ordering cost is Rs 75. The economic order quantity of the tyres is:

(A) 64
(B) 212
(C) 300
(D) 1200

Solution: (C)

D = 9600

C_o = Rs 75 per order

C_H = Rs 16 per tyre

$$EOQ = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 9600 \times 75}{16}} = 300$$

10. The annual demand for a product is 5000 units, and the ordering cost is Rs 7000 per order. Considering the basic economic order quantity model, the economic order quantity is 1000 units. When the annual inventory cost is minimised, the annual inventory holding cost (in Rs) is ___.

Solution:

D = 5000

C_o = Rs 7000 per order

EOQ = 10000

C_H = ?

$$EOQ = \sqrt{\frac{2DC_0}{C_H}}$$

Squaring on both sides

$$10000 \times 10000 = \frac{2 \times 50000 \times 7000}{CH}$$

C_H = 7 Rs/unit/year

11. The annual requirement of rivets at a ship manufacturing company is 2000 kg. the rivets are supplied in units of 1 kg costing Rs. 25 each. If it costs Rs. 100 to place an order and the annual cost of carrying one unit is 9% of its purchased cost, the cycle length of the order (in days) will be _____.

Solution:

D = 2000 kg

C = Rs 25 per kg

C_o = Rs 100 per order

$$C_H = \frac{29}{100} \times 25 \text{ Rs/Unit/Year}$$

$$EOQ = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 2000 \times 4 \times 100}{9}} = 421.6$$

$$N = \frac{D}{Q} = \frac{2000}{421 \cdot 6} = 4.74$$

1 year = 12 months

$$\frac{12}{4 \cdot 74} = 2.53 \text{ month}$$

$$= 2.53 \times 30 \text{ days}$$

$$= 75.9 \text{ days}$$

12. Consider the following data regarding the elementary deterministic economic order quantity model.

Annual demand for an item	100000
Unit price of the item (in Rs)	10
Inventory carrying cost per unit per year (in Rs)	1.5
Unit order cost (in Rs)	30

The total number of economic orders per year to meet the annual demand is _____.

Solution:

D = 10000
C = 10 Rs/kg
C_o = Rs 30/order
C_H = 1.5 Rs/Unit/Year

$$EOQ = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 100000 \times 30}{1.5}} = 2000$$

$$N = \frac{D}{Q} = \frac{100000}{2000} = 50$$

13. A manufacturer can produce 12000 bearings per day. The manufacturer received an order of 8000 bearings per day from a customer. The cost of holding a bearing in stock is Rs 0.20 per month. The setup cost per production run is Rs 500. Assuming 300 working days in a year, what should be the frequency of the production?

(A) 4.5 days
(B) 4.5 months
(C) 6.8 days
(D) 6.8 months

Solution: (C)

P = 12000/day
D = 8000/day
C_0 = 500
C_H = 0.20/unit/month

$$\sqrt{\frac{2DC_0P}{C_H(P \cdot D)}} = \sqrt{\frac{2 \times 8000 \times 300 \times 120 \times 500}{0 \cdot 20 \times 12 \times (12000.8000)}}$$

$$= 54772.25$$

$$N = \frac{D}{Q} = \frac{2400000}{54772.25} = 43.8$$

300/43.8 = 6.8 days

14. A company uses 2555 units of an item annually. The delivery lead time is eight days. What is the recorder point (in number of units) to achieve optimum inventory?

(A) 7
(B) 8
(C) 56
(D) 60

Solution: (C)

Q = ROL/LT
Q = 2555
LT = 8 days

$$= \frac{2555 \times 8}{365} = 56 \text{ ROL/year}$$

15. The maximum level of inventory of an item 100 and it is achieved with infinite replenishment rate. The inventory becomes zero over one and half month due to consumption at a uniform rate. This cycle continues throughout the year. Ordering cost is Rs 100 per order and inventory carrying cost is Rs 10 per item per month. Annual cost (in Rs) of the plan, neglecting material cost, is:

(A) 800
(B) 2800
(C) 4800
(D) 6800

Solution: (C)

Q = 100
Number of cycles = 12/1.5 = 8 cycles
C_0 = 100
C_H = 10/month = 120
TAC = P.C + O.C + HC

$$\frac{D}{Q} \cdot C_0 + \frac{Q}{2} C_H$$

$$= 8 \times 100 + \frac{100}{2} \times 120 = 6800$$

16. In a machine shop, pins of 15 mm diameter are produced at a rate of 1000 per month, and the same is consumed at a rate of 500 per month. The production and consumption continue simultaneously till the maximum inventory is reached. The inventory is allowed to reduce to zero due to consumption. The lot size of production is 1000. If the backlog is not allowed, the maximum inventory level is

(A) 400
(B) 500
(C) 600
(D) 700

Solution: (B)

P = 1000/month
d = 500/month
Q = 1000

$$I_m = Q\left(\frac{P-d}{P}\right) = 1000\left[\frac{1000-500}{1000}\right] = 500$$

17. A company's annual demand is 18000 units, and the cost price/unit is Rs 4. Ordering cost is C_o = 120/order and inventory carrying cost is 12% per annum of unit cost value. If lead time = 10 working days and total working days in a year are 300, then determine the following.

a. EOQ

b. Optimum number of orders/days

c. Length of the inventory cycle

Solution:

D = 18000

C = Rs 4 per kg

C_o = Rs 120 per order

$C_H = \frac{12}{100} \times 4$ Rs/Unit/Year

a. $EOQ = \sqrt{\frac{2DC_0}{C_H}} = \sqrt{\frac{2 \times 18000 \times 120 \times 25}{12}}$

= 3000

b. N = D/Q = 1800/3000 = 6

c. 300/6 = 50 days

18. P and Q are two products with the following characteristics as shown in the table. Find the ratio of EOQ of P and Q.

Product	(D) Demand (Unit/Year)	C_0 (Rs/ Order)	CH (Rs/ Unit/ Year)
P	100	200	4
Q	400	50	1

Solution:

$$\frac{P}{Q} = \frac{\sqrt{\frac{2 \times 100 \times 200}{4}}}{\sqrt{\frac{2 \times 400 \times 50}{1}}} = \frac{100}{200}$$

P : Q = 1 : 2

19. For a company D = 12000 units, production rate/month is 2000 units, and holding cost per unit per month is 0.15. If the unit cost is Rs 4 and set-up cost is Rs 400 per set-up, then find the optimum size.

Solution:

D = 12000 C = Rs 4

P = 2000/month C_0 = Rs 400

C_H = 0.15 × 12 = 1.8

d = D/T = 12000/12 = 1000/month

$$Q = \sqrt{\frac{2DC_0P}{C_H(P \cdot D)}} = \sqrt{\frac{2 \times 12000 \times 400 \times 2000}{1 \cdot 8 \times (2000 - 1000)}}$$

= 3265.98

≅ 3266

20. Consider the following data for an item. Annual demand = 2500 units per year, ordering cost = Rs 100 per order, and inventory holding rate = 25% of unit price. Price quoted by a supplier:

Order Quantity (Units)	Unit Price (Rs)
<500	10
>500	9

What is the optimum order quantity (in units)?

Solution:

D = 2500

C_o = 100/order

C_H = ¼ × c

$$EOQ = \sqrt{\frac{2 \times 2500 \times 100}{\frac{1}{4} * 10}} = 447$$

Q = 447, P = Rs 10

TAC = PC + OC + HC

$$447 \times 10 + \frac{2500}{447} \times 100 + \frac{447}{2} \times \frac{1}{4} \times 10$$

4470 + 559.2 + 558.75 = 5587.9

Q = 500, C = Rs 9

TAC = PC + OC + HC

$$5571 \quad 501 \times 9 + \frac{2500}{501} \times 100 + \frac{501}{2} \times \frac{1}{4} \times 9$$

Probabilistic Model

- In this model demand is uncertain and the decision is based on single ordering, i.e., Reordering is not permitted. In these models, we have to place a single order, and based upon this profit must be maximised.
- In this model, Factors encouraging higher safety stock are when demand rate and lead time variations are more, If the stockout cost is very high If the number of orders in a year

is more If the cost of holding inventory is very best and is not of much concerned.

ROL = Average Demand during lead time (ADDLT) + safety stock (SS)

ROL = LT × d + SS

$$\text{Average stock} = \frac{EOQ}{2} + SS = \frac{Q^*}{2} + SS$$

Inventory Management Techniques

Always Better Control (ABC)

- Annual usage value = Annual requirement × Per unit cost
- The ABC analysis is based on Pareto's law. In the ABC analysis, the items are classified into three main categories based on their respective usage value.
- Category A items are the costliest and the valuable items. 10% of items account for 75% of total capital invested in inventory. The items of this category should be ordered frequently but in small numbers.
    - A periodic review policy should be followed to minimise the shortage (%) of such items and top inventory staff should control these items. More effort is made to reduce lead time
    - Category B items have an average consumption value of nearly 15% of the items in an inventory account for 15% of the total investment.
    - These items have less importance than category A items, but they cannot be overlooked and require a lesser degree of control than those in category A.
    - Statistical sampling is generally useful to control them. A moderate effort is made to reduce lead time.
    - Category C items have low consumption value; nearly 75% of inventory items account only for 10% of the total invested capital.

VED Analysis

VED analysis is based on the criticality of the items, i.e., if the items are arranged in the descending order of their criticality.

V - Vital items

E - Essential items

D - Desirable items

- Vital items are those items that are not available when required. They make the entire system inoperative.
- Essential items are those that, if unavailable when requested, can diminish the system's efficiency.
- Desirable items are such that, even if they are unavailable, they do not prevent the system from functioning or lower its efficiency, but it would be preferable if they were available.

FNSD Analysis

FNSD analysis is particularly useful to combat obsolete items. The items can also be classified according to descending order, of their usage rate, or movement.

F - Fast moving items.

N - Normal moving items.

S - Slow moving items.

D - Dead items.

XYZ Analysis

This research enables us to discover the things that are heavily stocked. As we have seen, the consumption value of the commodities was the basis for ABC classification. However, the basis of the XYZ classification is the goods' closing Inventory value.

X - Items with high inventory value

Y - Items with moderate inventory value

Z - Items with low inventory value

GOLF Classification

GOLF classification of inventory items is done considering the nature of suppliers. The source of supply of different items are different, to determine the lead time, order quantities, safety stock, and terms of purchase and payment.

G - Government-controlled supplies

O - Open market supplies

L - Local supplies

F - Foreign market supplies

5 Reliability

Reliability

- Reliability can be defined as the probability that an item or system will perform its intended function without failure for a certain period under specified conditions such as operating and environmental conditions, etc.
- Reliability is changing of product or system quality over time, and it is a time-bound phenomenon
- If we have many items that we can test over time, then the Reliability of the items at time t is given by R = f(t)

  At time t = 0

  the number of survivors is equal to the number of items put on the test. Therefore, the reliability at t = 0 is

  R (t = 0) = 100% (initially without usage)

Aspects of Reliability

- Reliability is a probability-based concept, the numerical value of the reliability is between 0 and 1.
- The functional performance of the product must meet certain requirements and a functional definition of failure is needed.
- Reliability implies successful operation over a certain period.
- Operating or environmental conditions under which product usage takes place must be specified.

Design of Reliability

The following three key elements are required for the design of reliability.

- A commitment from top management to ensure reliability.
- A reliability policy that matches the quality policy of an organisation.
- A philosophy that designs reliability at an early stage of product design.

System Reliability

- Most items are composed of multiple components. The system's reliability (i.e., the reliability of the product) is determined by the component and system setup
- The reliability of a product is influenced by its design, manufacture, and maintenance; nevertheless, design is primarily responsible for its enhancement.
- Redundancy in design, which is typically accomplished by arranging components in parallel, is a standard method for boosting the system's reliability. So long as a single component is operational, the system will continue to function.
- Let us examine how to compute system dependability for systems with series, parallel, or both series and parallel components.

Systems with Components in Series



Fig. 5.1

- The diagram depicts a sequence of system components (C1, C2, C3, etc.). Each component must function for the system to function. Assume the components work independently of one another (i.e., the failure of one component does not influence the failure of any other component).
- If there are n components in a series, and the reliability of the nth component is indicated by Rn, then the system reliability is

  Rs = R1 * R2 * ..... * Rn
- The system's reliability reduces as the number of components in a series increases. When components are connected in series, the reliability is always less than or equal to that of the individual components.

  $R_S \leq \{R_1, R_2, ..., R_n\}$

- Although overdesigning each component improves reliability, its effect would be nullified by the number of components in series. In addition, manufacturing capabilities and resource constraints limit the maximum reliability of a specific component. Redesigning a product to reduce the number of series components is a possible approach.

Solved Examples

1. Find the reliability of the given system.

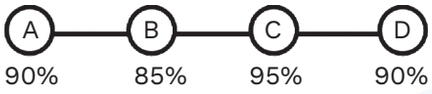


Fig. 5.2

Solution:

R_S : $R_A \times R_B \times R_C : 0.9 \times 0.85 \times 0.95 = 0.726$

2. Find the reliability of the given system.

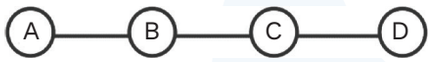


Fig. 5.3

$R_A = R_B = R_C = R_D = 1$

Solution:

$R_S = R_A \times R_B \times R_C \times R_D = 1$

Systems with Components in Parallel

- By arranging components in parallel, system reliability can be enhanced. The components are redundant; the system will continue to function as long as at least one component is operational. All parallel system components must fail simultaneously for the system to fail. An example of a system having parallel components (C1, C2, etc.). It is expected that all components operate simultaneously.

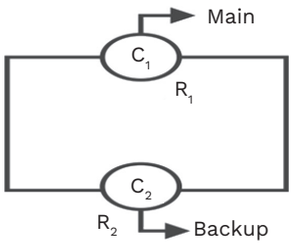


Fig. 5.4

- For example, the braking mechanism is a critical system in the automobile. Dual subsystems thus exist so that if one fails, the brakes still work.
- Assume we have n parallel components, with the reliability of the ith component given by Ri, where I = 1, 2, ..., n. Assuming random and independent operation of the components, the chance of failure of each component is given by Fi = 1 – Ri. Now, the system only fails if every component fails. Consequently, the likelihood of system failure

Fs = (1 – R1) (1 – R2) ... (1 – Rn)

Solved Examples

3. Find the reliability of the given system.

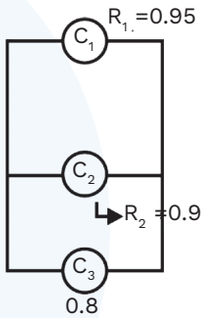


Fig. 5.5

Solution:

$(\bar{R}_1) = 1 - 0.95 = 0.05$

Similarly, we can write the failure rate for C2 and C3

$R_S = 1 - (0.05)(0.1)(0.2)$

4. Find the reliability of the given system.

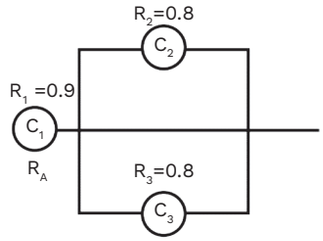


Fig. 5.6

Solution:

$R_B = 1 - (1 - 0.8)(1 - 0.8) = 0.96$

$R_S = R_A \times R_B = 0.9 \times 0.96 = 0.864$

Mean Time to Failure (MTTF)

MTTF applies to non-repairable items or devices and is defined as the average time an item may be expected to function before failure. This can be estimated from a suitable sample of items, which have been tested to the point of failure. MTTF is simply the average of all the times to failure.
For example, four items have lasted 3000 hours, 4000 hours, 4000 hours, and 5000 hours.

$$\frac{3000 + 4000 + 4000 + 5000}{4}$$

MTTF is 4000 hours.

Mean Time to Repair (MTTR)

MTTR is estimated by dividing the cumulative repair time, in a given elapsed time, by the cumulative number of repairs (or failures) of the component in the same elapsed time, i.e.,

$$MTTR = (t, + t_2 + \ldots + t_n)/n$$

t = repair time for the ith failure
n = number of failures for which repair times were available

Mean Time between Failure (MTBF)

- MTBF applies to repairable items. MTBF refers to the time between failures for obvious reasons. MTBF can be calculated as

  $$\frac{\text{Total device hours}}{\text{Number of failures}}$$

- Let us consider an item that has failed, say, 4 times for 16000 hours. Then MTBF is 16000/4 = 4000 hours. (This is, of course, just the same method as for MTTF).
  The mean time to failure (MTTF) for the exponential distribution is given as

  $\lambda = 1/MTTF$

  λ is the failure rate.

Failure Rate (λ): The rate at which failures occur.

- It is typically measured in terms of the number of failures per unit time of components exposed to conditions under which failure can occur.
  If n = number of failures of a component over a period in which it operated for time 't'.
  Then, an estimate of the failure rate (λ) is given by

  $\lambda = n / t$

- For repairable equipment, this is also equal to the mean time between failures (MTBF). There will be a difference between MTBF and MTTF only if there is a significant repair or replacement time upon failure of the product.

  MTBF (or MTTF) = $1/\lambda$

- For continuously operating repairable systems/components, the relation between MTBF, MTTR, and MTTF is

  MTBF = MTTF + MTTR

Reliability Prediction Using Exponential Distribution

- The exponential distribution applies when the failure rate is constant—the graph is a straight horizontal line, instead of a 'bathtub'.
- It is used to predict the probability of survival at a particular time. If λ is the failure rate and t is the time, then the reliability, R, can be determined by the following equation:

  $$R(t) = e^{-\lambda t}$$

Solved Examples

5. Find the reliability of the given system.



Fig. 5.7

Solution:

$R_1 = e^{-\lambda_1 t}$ $\quad R_2 = e^{-\lambda_2 t}$

$R_S = R_1 \times R_2$

$R_3 = e^{-\lambda_1 t} \times e^{-\lambda_2 t} = e^{-(\lambda_1 + \lambda_2)t}$

$\lambda_5 = \lambda_1 + \lambda_2 + \ldots + \lambda_n$

If $\lambda_1 = \lambda_2 = \ldots = \lambda_n$

$\lambda_S = n\lambda$

$$MTTF = \frac{1}{\lambda_S} = \frac{1}{n\lambda}$$

6. Find the reliability of the given system.

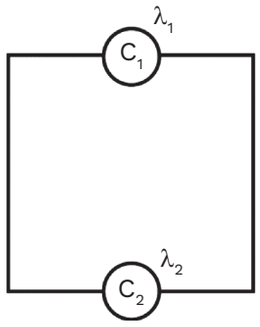


Fig. 5.8

Solution:

$(\bar{R}) = 1 - R_1 = 1 - e^{-\lambda.t}$

$R_S = 1 - (1 - e^{-\lambda,t})(1 - e^{-\lambda_2 t})$

MTTF of the system in parallel connection

$$\frac{1}{\lambda} = \left[1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{n}\right]$$

7. An acid central system has three components in series with individual reliabilities (R_1, R_2, R_3) as shown in the figure.

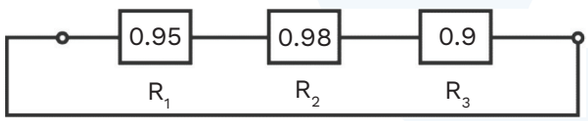


Fig. 5.9

If a parallel system similar to the above system is designed, then what is the reliability of the entire system?

(A) 0.257
(B) 0.587
(C) 0.796
(D) 0.973

Solution: (D)

R = R1*R2*R3

0.95*0.98*0.9 = 0.8379

Rs = 1 – (1 – Ra) (1 – Rb)

1 – (1 – 0.8379) (1 – 0.8379)

0.973

8. A system contains three components. The reliability of each component is measured for 900 hours, and the time to failure components are found to be exponential. If each component has a reliability of 0.9, then what will be the system reliability at the end of 900 hours? [The system is connected in series].

(A) 0.729
(B) 0.587
(C) 0.296
(D) 0.973

Solution: (A)

$R_S = R_1 \times R_2 \times R_3 = (0.9)^3 = 0.729$

9. The reliability block diagram of a system is shown in the following figure with the component reliability rated in each block:

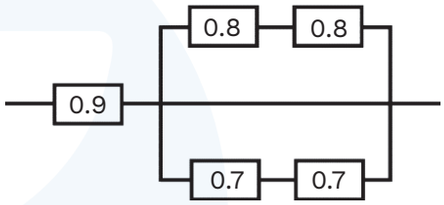


Fig. 5.10

What is the reliability of the given system?

(A) 0.74
(B) 0.8
(C) 0.74
(D) 0.54

Solution: (A)

$R_A = 0.64; \bar{R}_A = 1 - 0.64 = 0.36$

$R_B = 0.49 = \bar{R}_B = 1 - 0.49 = 0.51$

$R_1 = 1 - (\bar{R}_A)(\bar{R}_B)$

$1 - (0.36)(0.51)$

0.8164

$R_S = R_1 \times R_2 = 0.8164 \times 0.9 = 0.734$

10. Consider the following aspects of reliability

I. Reliability is a probability-based concept, and the numerical value of the reliability is between 0 and 1.
II. Reliability is a measure of the result of the quality of the product over the long run.
III. Reliability is an engineering discipline for applying scientific know-how to a

component, assembly, plant, or process, so it will perform, its intended function, without failure, for the required time duration when operated correctly in a specified environment

Which of the above statements are correct?

(A) I and II
(B) II and III
(C) I and III
(D) I, II, and III

Solution: (D)

11. The reliability of equipment for a time to failure exceeding t hours is given by $R(t) = e^{-0.005t}$.

In the meantime, to failure (MTTF) for this equipment (in hours) is:

(A) 100
(B) 400
(C) 200
(D) None

Solution: (C)

$\lambda = 0.005$

$$MTTF = \frac{1}{\lambda}$$

$$\frac{1}{5 \times 10^{-3}} = \frac{1000}{5} = 200 \text{ hours}$$

12. An amplifier has an exponential time to failure distribution with a failure rate of 8% per 1000 hours. What is the reliability of the amplifier at 5000 hours? Find the meantime to failure.

Solution:

$$\lambda = \frac{0.08}{1000} = 0.00008/\text{hour}$$

$$R(t = 500) = e^{-\lambda t}$$

$$e^{-0.00008 \times 5000}$$

$$e^{-0.4} = 0.6703$$

$$MTTF = \frac{1}{\lambda} = \frac{1}{0.00008} = 0.125 \times 10^5$$

$$= 12500 \text{ hours}$$

13. What is the highest failure rate for a product if it is to have a 95% likelihood of survival (i.e., successful operation) at 4000 hours? Assume an exponential distribution for the time to failure.

Solution:

$$R = e^{-\lambda t}$$

$$0.95 = e^{-\lambda \times 4000}$$

$$\lambda = 0.0000128/\text{hour} = 12.8 \times 10^{-6}/\text{hour}$$

14. A module of a satellite monitoring system has 500 components in series. The reliability of each component in series is reduced to 200, what is the reliability of the module?

Solution:

$$R_S = R_1 \times R_2 \times \ldots \times R_N$$

$$R_s = (0.999)^{500} = 0.606 = 60.6\%$$

$$R_s = (0.999)^{200} = 0.819 = 81.9\%$$

15. The television camera's automatic focus unit consists of ten components arranged in series. Each component's time–to–failure distribution is exponential, with a constant failure rate of 0.05 per 4000 hours. How reliable is each component following 2000 hours of operation? Determine the unit's reliability after 2000 hours of use. What is its mean time for failure?

Solution:

$$\lambda = \frac{0.05}{4000} = 12.5 \times 10^{-6}/\text{hour}$$

$$R(t - 2000), e^{-\lambda t} = e^{-12.5 \times 10^{-6} \times 2000} = 0.975$$

$$R_S = e^{\frac{-(\lambda_1 + \lambda_2 + \ldots + \lambda_n)t}{n\lambda}}$$

$$e^{-10 \times 12.5 \times 10^{-6} \times 2000} = 0.779$$

$$MTTF = \frac{1}{n\lambda}$$

$$\frac{1}{10 \times 12.5 \times 10^{-6}}$$

8000 hours

16. Refer to the question above concerning the automatic focus unit of a television camera which has 10 similar components in series. It is desired for the focus unit to have a reliability of 0.95 after 2000 hours of operation. What would be the meantime to failure of the individual components?

Solution:

$R_S = e^{-(\lambda_1+\lambda_2+...+\lambda_n)t}$

$0.95 = e^{-\lambda_S \times 2000}$

$\lambda_S = 0.0000256 / \text{hour}$

$\lambda_5 = n\lambda \quad \lambda = \frac{\lambda_S}{10}$

$2.56 \times 10^{-6} / \text{hour}$

$\text{MTTF} = \frac{1}{\lambda} = \frac{1}{2.56 \times 10^{-6}} = 390625 \text{ hours}$

17. Find the reliability of the system shown in the figure with three components (A, B, and C) in parallel. The reliabilities of A, B, and C are 0.95, 0.92, and 0.90, respectively.

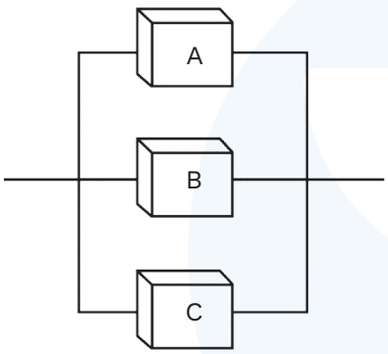


Fig. 5.11

Solution:

$R_S = 1-(1-0.95)(1-0.92)(1-0.90)$

$1-0.0004 = 0.996$

18. For the system shown in the figure determine the system reliability for 2000 hours of operation and find the meantime to failure. Assume that all three components have an identical time–to–exponential failure distribution, with a constant failure rate of 0.0005/hour. What is the meantime for the failure of each component? If it is desired for the system to have a mean time to failure of 4000 hours, what should the meantime to failure be for each component?

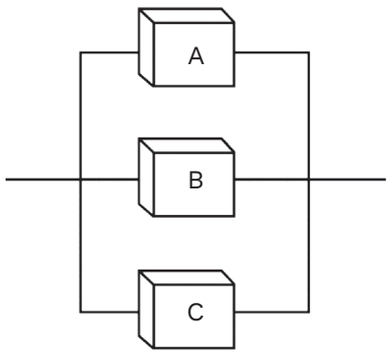


Fig. 5.12

Solution:

$\lambda = 0.0005 / \text{hour}$

$R_S = 1-[1-e^{-\lambda t}]^3$

$1-[1-e^{-0.0005 \times 2000}]^3$

$1-[0.692.2]^3$

0.7474

$\text{MTTF} = \frac{1}{\lambda}$

$\frac{1}{0.0005} = 2000 \text{ hours}$

$\text{MTTF} = \frac{1}{\lambda}\left[1+\frac{1}{2}+\frac{1}{3}\right]$

$\frac{1}{0.0005}\left[\frac{6+3+2}{6}\right]$

3666.6 hours

MTTFs = 4000

$\text{MTTFs} = \frac{1}{\lambda}\left[1+\frac{1}{2}+\frac{1}{3}\right] \rightarrow 0.00046 / \text{hour}$

$\text{MTTF} = \frac{1}{\lambda} = \frac{1}{0.00046} = 2173.91 \text{ hours}$

19. Find the reliability of the eight–component system shown in figure some components are in series and some are in parallel. The reliabilities of the components are as follows: R_A = 0.92, R_{A2} = 0.90, R_{A3} = 0.88, R_{A4} = 0.96, R_{B1} = 0.95, R_{B2} = 0.90, R_{B3} = 0.92, and R_{C1} = 0.93

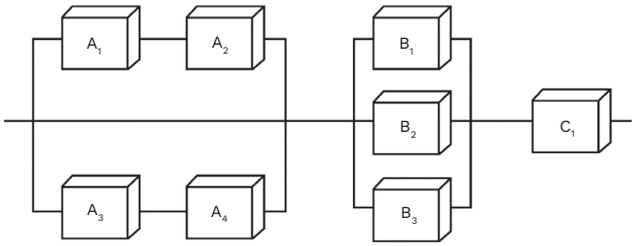


Fig. 5.13

Solution:

$R_n = 1-(1-0.828)(1-0.8448)$

$R_A = 0.9733$

$R_S = R_A \times R_B \times R_C$

$0.9733 \times 0.9996 \times 0.93$

$R_S = 0.9048$

20. Find the system failure rate and the meantime to failure for the eight-component system shown in Figures 11 – 6. The failure rates (number of units per hour) for the components are as follows;

$\lambda_{A_1} = 0.0006$, $\lambda_{A_2} = 0.0045$, $\lambda_{A_3} = 0.0035$, $\lambda_{A_4} = 0.0016$, $\lambda_{B1} = 0.0060$, $\lambda_{B2} = 0.0060$, $\lambda_{B3} = 0.0060$, $\lambda_{c1} = 0.0050$.

Solution:

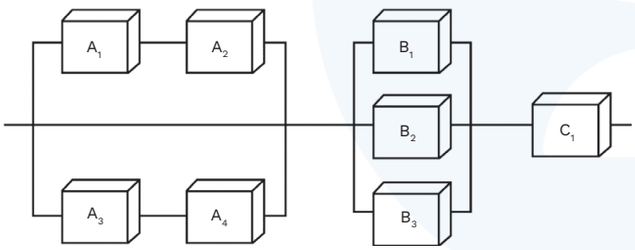


Fig. 5.14

$\lambda_S = 0.0035 + 0.0016 = 0.0051$

$$\text{MTTF} = \frac{1}{\lambda}\left[1+\frac{1}{2}+\frac{1}{3}\right]$$

$$\text{MTTF} = \frac{1}{\lambda}\left[1+\frac{1}{2}\right]$$

$$\frac{1}{0.0051}\left[\frac{3}{2}\right] = 294.11 \text{ hours}$$

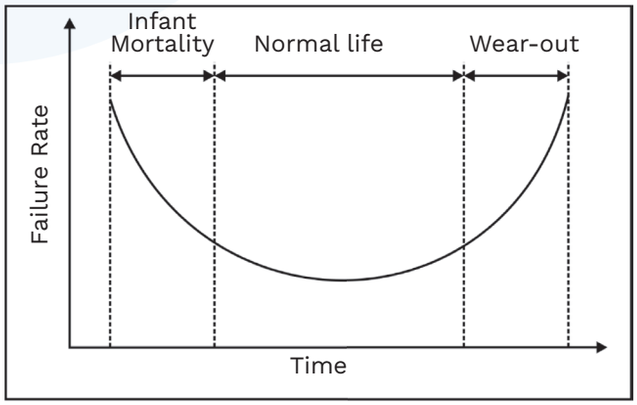
$$\frac{1}{0.006}\left[\frac{11}{6}\right] = 305.55 \text{ hours}$$

$$\lambda_S = \frac{1}{294.11}+\frac{1}{305.55}+0.005 = 0.01167$$

$$\text{MTTF} = \frac{1}{\lambda_5} = \frac{1}{0.01167} = 85.6 \text{ hours}$$

Bathtub Curve/Life Cycle Curve

- A bathtub curve is a graphical representation of a product population's lifetime. The bathtub curve does not represent the failure rate of a single item, but rather the failure rate of a population of products over time.
- The bathtub curve reflects the failure pattern of numerous products, particularly sophisticated ones such as automobiles and washing machines. The vertical axis of the graph represents the rate of failure at each time point. Higher levels suggest a greater likelihood of failure.
  - Failure happens when stress surpasses strength due to flaws, randomly greater loads, or wear out/fatigue.
  - From product conception through product demise, the majority of products pass through three distinct phases. The graph below depicts a typical life-cycle curve representing the instantaneous failure rate from birth to death as a function of time.
  - The bathtub curve is separated into three regions: infant mortality, useful life, and wear-out stage.



Bathtub Curve

Fig. 5.15

Infant Mortality

This phase is also known as the early failure or debugging phase. The rate of failure is considerable but diminishes significantly over time. During this period, failures are caused by defective items and testing. Consequently, the failure rate at the beginning of the infant mortality stage is high and subsequently lowers over time as early failures are eliminated by burn-in or other stress screening measures.

Useful Life

The middle phase of the bathtub curve is also known as the chance failure phase. This phase is marked by a steady rate of failure. This period receives the greatest consideration throughout the design phase and is the most important for reliability prediction and evaluation activities.

Wear-Out Stage

This is the final phase in which the failure rate rises as the products begin to deteriorate due to age or lack of maintenance. When the failure rate becomes excessive, repair, replacement of components, etc., should be performed.

Maintenance

- Maintenance is defined as "the combination of all technical and corresponding administrative actions intended to retain an item in or restore it to a state in which it can perform its required function.
- Maintenance is a result of design, and it includes servicing, removal and replacement, inspection, calibration, overhaul, and condition verification.
- Maintenance aims to maintain a high level of equipment reliability and maintainability as contributors to production/operations, maximise economy in equipment management for its entire life, cultivate equipment-related expertise and skills amongst maintenance personnel, and create a vigorous and enthusiastic work environment.

Maintainability

- It is the ability of an item under given conditions of use, to be retained in or restored to a state in which it can perform the required function. Hence, maintenance is performed under given conditions using stated procedures and resources.
- Maintainability may be measured in terms of the expected amount of time required to restore operability after a failure or in terms of the probability of restoring operability (after a failure) in a given period of time.
- The improvement in maintainability is an important aspect of the design and maintenance department. The various policies and decisions related to maintainability can be classified as a design decision, maintenance policies, technician requirement
- The combined effect of these three affects the various components of downtime viz. detection time, diagnosis time, correction time, and verification time.

Availability

- The ability of an item, to be in a state to perform required functions at any instant of time or during a stated interval of time assuming the required external resources are provided.
- The Fraction of a given period during which an item was available to perform required functions is a commonly used measure of Availability. Attaining the desired level of system availability requires a complex process involving many resources of which system reliability and maintainability requirements are generalised characteristics describing system performance during a period. The system availability A can be described as follows, assuming steady-state conditions.

$$A = [MTBF/(MTBF + MTTR)]$$

A = availability of the system
MTBF = mean time between failure
MTTR = mean time to repair
Availability can also be expressed as the input-output function

$$A = f(R, M, S)$$

A Availability
R Reliability
M Maintainability
S Supply effectiveness

Availability can also be defined as Operating time/operating time + Downtime.

- Downtime may consist of active repair time, administrative time (processing of necessary paperwork), and logistic time (waiting time due to lack of parts).
- In the situation when the time-to-failure distribution is exponential (with a failure rate λ) and the time-to-repair distribution is also exponential (with a repair rate μ), the availability is given by $\mu/(\lambda + \mu)$.

Reactive/Breakdown Maintenance

Reactive maintenance is the maintenance strategy of running it until it breaks. No measures or attempts are performed to maintain the equipment in accordance with the designer's original intent to achieve the design life. In short, immediate replacement or repair when a problem occurs.

Preventive Maintenance

Preventive maintenance is operations performed on a time- or machine-run-based schedule that detect, prevent, or mitigate degradation of a component or system to sustain or prolong its useful life by regulating degradation to an acceptable level. Preventive maintenance's primary objective is to preserve and improve equipment dependability.

Predictive Maintenance

- Predictive maintenance can be described as measurements that detect the onset of system degradation (lower functional state), allowing the causal stress to be eliminated or regulated before the component's physical state degrades significantly. The outcomes reveal present and future functional capabilities. Predictive maintenance's major objective is to minimise disruptions to normal system operations while permitting budgeted, scheduled repairs.
- Predictive maintenance differs from preventive maintenance based on maintenance needs on the actual condition of the machine rather than on some pre-set schedule. Preventive maintenance is time-based.

Condition Monitoring (CM)

- Condition monitoring is widely used to ascertain the health of machinery, equipment, vehicles, and other valuable assets. It provides crucial information that ensures equipment is operating at optimum efficiency, minimizing downtime, and reducing defects. Condition monitoring has been useful for understanding the historical performance and status of equipment, but it does not do a great job of preventing or mitigating potential.
- Condition monitoring is not something new. It has been done by Maintenance personnel, who kept their eyes and ears open when they went through the working area. But the development of technology has replaced these human sensors with sophisticated equipment which are essential for higher fault detection efficiently.

Classification of Condition Monitoring

Depending on the time of monitoring of various parameters, condition monitoring can be classified into two types.

- On-load monitoring techniques
- Off-load monitoring techniques

Load Monitoring Techniques

On-load monitoring techniques refer to the monitoring scheme performed during the operation of a system. The various schemes used under load monitoring schemes include visual inspection of accessible components, temperature monitoring, lubrication monitoring, leak detection, vibration monitoring, and noise monitoring.

Offload Monitoring Techniques

- Offload monitoring techniques refers to the monitoring scheme performed during the non-operation period of a system. The various schemes used under offload monitoring schemes include the period of visual inspection of inaccessible or moving parts, crack detection, leak detection, and vibration testing.
- The most common techniques used for condition monitoring are thermal monitoring, lubricant monitoring, vibration monitoring, and leak monitoring.

- In lubricant monitoring techniques, lubricant is used to reduce the friction between two sliding surfaces, lubricant monitoring is done by debris deposited, debris in suspension, and condition of used oil.
- The debris deposited and in suspension is monitored using filters and magnetic debris collectors.
- Filters are used to determine the element content in the debris, and the size and shape of the debris. In case any deviation is there from the normal value they can give information about the fault that can occur.
- Magnetic debris collectors are used for capturing ferrous components and using magnetic plugs readily removable on-load and one magnetic debris collector for monitoring without removal.
- Thermal monitoring is used to detect an increase in heat generation due to some malfunction such as a damaged bearing. For thermal monitoring, there are two types of sensors used known contact sensors and non-contact sensors.
- Contact type sensors make direct contact with the surface. Commonly used contact type sensors are liquid expansion sensors, bimetallic expansion sensors, thermocouple sensors, and resistance sensors.
- Non-contact type sensors use infrared technology. They scan the temperature profile of the focused area. After scanning, they generate a coloured thermal image of the area being focused on the viewfinder. The colour profile indicates the temperature profile of the area. Commonly used non-contact type sensors are optical pyrometer, radiator pyrometer, infrared scanner.
- All machine systems vibrate as vibration is one of the side effects of machine operation. Vibration is defined as the motion of a machine component back and forth from its position of rest under a given set of conditions.
- Vibration is the behaviour of mechanical components as they react to internal or external forces. Vibration analysis traditionally has been one of the most complex and expensive predictive maintenance tools. Interpreting vibration spectra is relatively subjective, and it is difficult to set effective alarm limits for the rotating machine.
- In vibration analysis, vibration data available from various sensors is processed for measuring, continuously monitoring, trending, comparing, and analysing for corrective action to be taken on various machines to prolong their working conditions. From the measured parameter, we can have the information as frequency origination, amplitude severity, and phase relative position.
- Reliability-centred maintenance (RCM) is the ongoing, systematic process of matching critical systems with the most cost-effective maintenance strategy to maximise overall reliability.
- The RCM technique addresses major challenges that conventional maintenance plans do not address. It acknowledges that not all equipment within a facility is of similar significance to the process or facility safety. It acknowledges that the design and operation of equipment vary and that varied equipment has a greater likelihood of failing due to distinct mechanisms of deterioration. In addition, it recognises that a facility's financial and human resources are finite and that their utilisation must be prioritised and optimised while building a maintenance programme.

Performance Indicators for Maintenance

Based on equipment performance

- Availability = Operating time/Operating time + Downtime
- Mean time between failures (MTBF) = (Number of operating hours)/(Number of failures or breakdowns)
- Mean time to repair (MTTR) = (Sum of repair time)/Number of breakdowns
- Number of plant outages caused due to equipment failure
- Number of breakdowns during the reported period
- Number of repetitive failures during the reported period

Based on work control

- Work control indicator (WCI) = Number of maintenance work orders received each math/Number of maintenance work order closed during month
- Preventive maintenance indicators = Number of preventive maintenance jobs planned – Number of preventive maintenance jobs completed

6 Line Balancing

- The objective of line balancing is to group the facility or task efficiently to achieve the optimal balance of capacity or production system.
- An assembly line consists of a series of progressive assembly stations connected by material-handling machines. An assembly line is a unique form of product layout in which operations at several stations apply to the assembling of various parts.
- Line of product layout is advantageous for high-volume, single-product manufacturing operations. A moving conveyor may bring the work unit or sub-assembly units close to the employees, who then transport them to the subsequent station and perform the necessary activities. At each station, one or more employees do the necessary tasks.
- In an assembly line, the design of the workstation poses a challenge. Each workstation is intended for a limited number of processing and assembly duties. The goal of the design is to assign processes and activities to specific workstations so that the total time required at each workstation is roughly equivalent and closer to the intended cycle time or production rate.
- A case of perfect line balancing exists when all work elements that can be grouped at any station have the same station time. In this instance, production flow would be smooth. However, this is difficult to accomplish in practice When complete line balance is not accomplished, the output rate or cycle time is determined by the station time of the slowest station.

Basic Terminologies of Line Balancing

Work Element (i)

- The job is divided into its component tasks so that the work may be spread along the line. The work element is a part of the total job content in the line. Let N be the maximum number of work elements, which is obtained by dividing the total work into minimum rational work elements.
- The minimum rational work element is the smallest practical divisible task into which a work can be divided. Thus, the work element number (i) is $1 \le i \le N$. The time in a work element, I say (T_{iN}), is assumed as constant.

Workstation (Ws)

It is a location on the assembly line where a combination of a few work elements is performed. Since the minimum number of workstations (Ws) cannot be less than 1, we have

$Ws \ge 1$

Total Work Content (T_{wc})

This is the algebraic sum of time of all the work elements on the line. Thus

$$T_{wc} = \sum_{i=1}^{N} T_{iN}$$

Station Time (Tsi)

It is the sum of all the work elements (i) on the workstation. Thus, if there are n_1 to n_2 work elements assigned at stations, then.

$$T_{si} = \sum_{n_1}^{n_2} T_{iN}$$

Cycle Time (Tc)

Cycle time is the rate of production. This is the time between two successive assemblies coming out of a line. Cycle time can be greater than or equal to the maximum of all times, taken at any station

$Tc \ge (T_{si})$

If, T_c = Tmax (T_{si}), then there will be an ideal time at all stations having stations timeless than the cycle time.

Delay or Idle Time at Station (T_{ds})

This is the difference between the cycle time of the line and station time.

$T_{ds} = Tc - T_{si}$

Precedence Diagram

This is a diagram in which the work elements are shown as per their sequence relations. Any job cannot be performed unless its predecessor is completed.

Balance Delay (BD)

This is a measure of line inefficiency. Therefore, the effort is done to minimise the balance delay. Due to the imperfect allocation of work elements along with various stations, there is an idle time at the station. Therefore

$$\text{Balance delay} = \frac{nT_c - T_{wc}}{nT_c} = \frac{nT_c - \sum_{i=1}^{n} T_{iN}}{nT_c}$$

T_c = Total cycle time
T_{wc} = Total work content
n = Total number of stations

Line Efficiency

It is expressed as the ratio of the total station time to the cycle time, multiplied by the number of stations (n).

$$\text{Line Efficiency} = \frac{\sum_{i=1}^{n} T_{si}}{(n)(T_c)} \times 100\%$$

or
1 – BD

Smoothness Index (SI)

The smoothness index is a measure of the relative smoothness of a line.

$$SI = \sqrt{\sum_{i=1}^{n} \left[(T_{si})max - T_{si}\right]^2}$$

Solved Examples

1. A work shift is for 8 hours duration: 30 minutes lunch break and two 15 minutes (each) tea breaks are allowed per shift. If products are to go out after assembly at the rate of 60 per shift, and the total assembly time content for a product is 42 minutes, then the minimum number of workstations needed is:

(A) 8
(B) 12
(C) 6
(D) 5

Solution: (C)

Total Duration = 8 hours
Idle Time = 1 hour
Active time = 7 hours
1 product – 42 mins
60 products – 60 hours
Twc – 60 hours
Minimum number of workstations required = 42 hours/7 = 6 workstations

2. In an assembly line for assembling toys, five workers are assigned tasks that take times of 10, 8, 6, 9, and 10 minutes respectively. What is the balance delay for the line.

Solution:

$T_C \geq [T_{Si}]_{max} \rightarrow 10\,min$

$T\omega c = 10 + 8 + 6 + 9 + 10 = 43$

$n = 5$

$$BD = \frac{nT_C - T\omega_C}{nT_C} = \frac{50 - 43}{50} = \frac{7}{50} \times 100$$

BD = 14%
Line efficiency = 1 – BD = 86%

3. The table gives details of an assembly line. What is the smoothness index of the assembly line?

Workstation	I	II	III	IV	V	VI
Workstation (In Minutes)	7	9	7	10	9	6

Solution:

n = 6

$T_C \geq [T_{Si}]_{max}$

$T_{\omega c} = 48$

$$BD = \frac{60 - 48}{60} = \frac{12}{60} \times 100 = 20\%$$

Line efficiency $= 1 - BD = 80\%$

$$SI = \sqrt{\sum_{i=1}^{n}\left[(T_{si})max - T_{si}\right]^2}$$

$$\sqrt{(10.7)^2 + (10.9)^2 + (10-7)^2 + (10-10)^2 + (10.9)^2 + (10.6)^2}$$

$$\sqrt{9+1+9+0+1+16} = 6$$

4. Match the following failure with their respective reason in the bathtub curve.

P. Infant mortality	I. Due to stress exceeding strength
Q. Useful life	II. Due to fatigue of material depletion
R. Wear out design	III. Due to inherent defects and poor

	P	Q	R
(A)	II	III	I
(B)	I	II	III
(C)	III	I	II
(D)	III	II	I

Solution: (C)

5. Which of the following statements are true about assembly line balancing?

I. As the cycle time increases, the output of the line decreases.

II. As the cycle time increases, the number of workstations required decreases.

(A) I only
(B) II only
(C) Both I and II
(D) None of these

Solution: (A)

7 Sampling and OC Curves

Sampling

- It is a procedure in which random samples are drawn; (without showing any favour or bias). To ensure perfect randomness in sample selection, the type and configuration of the inspected parts may influence the steps or procedures that must be enacted to achieve this.
- Due to a lack of time and resources, obtaining data on a specific quality feature for each item in a population is sometimes impractical in quality control. In addition, sample data provide sufficient information on a product or process characteristic for a fraction of the expense. Therefore, it is essential to understand how samples are picked and the characteristics of various sampling processes.
- A sampling design is a description of the technique for selecting observations for a sample. It is not necessarily about the measuring instrument to be employed. For instance, a sampling plan may include selecting every tenth item manufactured
- An element is an object (or set of things) for which data or information is acquired in the context of sampling. A sampling unit is a single element or a collection of components from a population that does not overlap. A sampling frame is an account of all sampling units.
- In the sampling inspection, for a lot size N, we have to first determine Sample size (n) then the acceptance number, C
- Now each item in the sample is inspected.
    - If the number of defects in the sample is greater than the acceptance number, C, then reject the lot.
    - If the number of defects in the sample is less than the acceptance number, C then the whole lot is accepted.
- The only way to achieve 100% acceptable quality is to use 100% inspection.
- The advantage of 100% inspection is that the probability that the batch will be accepted is 1.0, if its quality is equal to or greater than the acceptance quality level (AQL) and zero if the quality is lower than the AQL.
- The term ‘acceptance quality level’ has less
- meant in 100% inspection, since a target of zero defects should be attainable if each part in the batch is inspected.
- Theoretically, 100% inspection allows only good quality to pass through an inspection process. However, when 100% inspection is done manually, two problems arise. First, the inspection cost sometimes exceeds the cost of making the part. Second, there are always errors associated with 100% inspection (type I and II errors) when the inspection procedure is performed by human inspectors. Because of these errors, 100% inspection using manual methods has no guarantee of 100% good quality product.

Acceptance Sampling

- It was popularised by Dodge and Roming during World War II and originally applied by the US military for the testing of bullets during World War II.
- Acceptance sampling is an important field of statistical quality control, where a random sample is taken from a lot, and upon the results of appraising the sampling, the lot will either be rejected or accepted.
- It is also a procedure for sentencing incoming batches or lots of items without doing 100% inspection. So, acceptance sampling is also known as ‘the middle of the road’ approach between no inspection and 100% inspection. The costs associated with 100% inspection are prohibitive, and the risks associated with 0% inspection are very large.
- In 1969, Harold Dodge noted that acceptance quality control is distinct from acceptance sampling, as the latter is dependent on specific sampling plans, which, when implemented, indicate the conditions for acceptance or rejection of the immediate lot being implemented, whereas the former can

be implemented in the form of an acceptance control chart.

- It is the form of inspection applied to lots or batches of items before or after a process, to judge conformance with predetermined standards.
- It is important to notice that acceptance sampling gives no idea about the process that is producing these items.

Acceptance Sampling by Attributes

It is based on 'go' and 'no-go' gauge inspection. Each item tested is classified as conforming or nonconforming. A sample is taken, and if it contains too many nonconforming items, the batch is rejected, otherwise it is accepted.

Acceptance Sampling by Variables

- It is also known as continuous measurement, usually measured by mean and standard deviation. Variables such as thickness, strength, or weight can be measured.
- It is easier to classify an item as conforming or non-confirming than to make an exact measurement. However, the information gained from an exact measurement is greater and so smaller sample sizes are required
- A decision as to whether to use acceptance sampling by attributes or variables will depend on the circumstances of each case.

Lot Selection

- Lot selection is also critical for the sampling inspection, and the lot should be produced on the same machinery, by the same operator, using the same raw materials, at roughly the same time.
- Larger quantities are superior to lesser quantities since they are more indicative of the overall quality. Lots must be compatible with the people and material handling systems.

Sampling Plan

- Choosing a suitable sampling plan that specifies lot size, sampling size, number of samples, and acceptance/rejection criteria is the most critical aspect of acceptance sampling.
- The sample size has a direct effect on the accuracy of the data's contents. The data are more valuable with larger sample size. Typically, it is of interest to determine the smallest sample size that can be used to estimate an unknown product or process parameter with acceptable precision, given a range of acceptable tolerance levels.
- There are three types of acceptance sampling plans.
  - Single sampling plan
  - Double sampling plan
  - Multiple sampling plan

Single Sampling Plan

- When a decision for acceptance or rejection of the lot is made based on only one sample, the acceptance plan is known as a single sampling plan.
- It is the most common and easiest plan to use but not the most efficient in terms of the average number of samples needed.
  In this, one sample was drawn from the lot and inspected.
  Let, N = Lot size
  m = Sample size (randomised)
  C = Acceptance number
  d = number of defective items in a sample of size n
  Rule: If d < C, accept the lot, else reject the lot.
- The advantages of a single sampling plan are less expensive because of less inspection and work with a single sample, while the disadvantages are the risk of accepting a lot of poor quality and risk of rejecting a lot of acceptable quality.

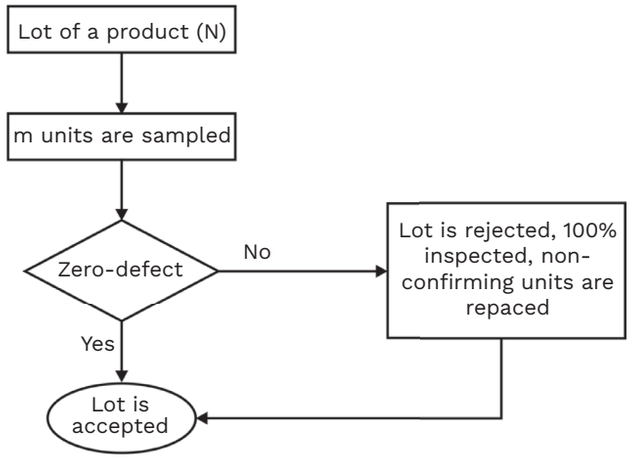


Fig. 7.1

Double Sampling Plan

- In a double sampling plan, the decision on acceptance or rejection of the lot is based on two samples.
- In the double sampling plan, first, take a small initial sample
  If the number of defectives < lower limit, accept the lot
  If the number of defectives > the upper limit, reject the lot
  If the number of defectives is between limits, then take a second sample
  Let m = Lot size
  n_1 = Number of pieces in the first sample
  C_1 = Acceptance number for the first sample
  n_2 = Number of pieces in the second sample
  $n_1 + n_2$ = Number of pieces in the two-sample combined
  C_2 = Acceptance number for the two samples combined

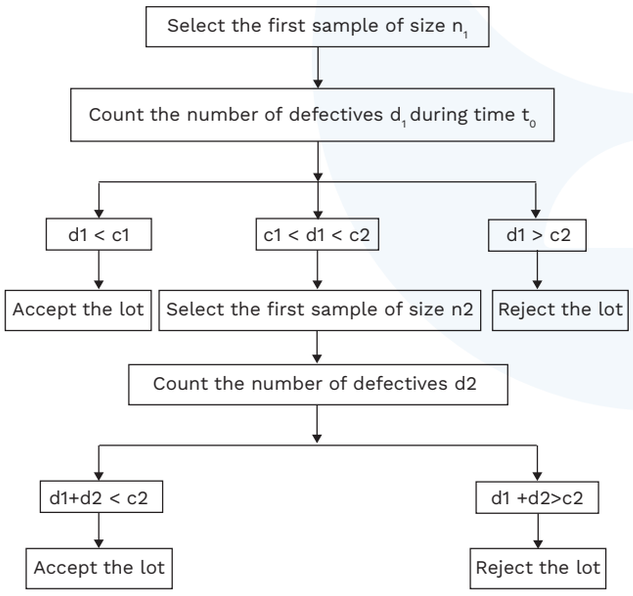


Fig. 7.2

Multiple Sampling Plan

- In this plan, a decision to accept or reject a lot is taken after inspecting more than two samples. In case, the cumulative number of defectives in a sample is more than the upper limit specified for that sample, the sampling procedure is terminated, and the lot is rejected. On the other hand, if the total number of defectives is less than or equal to the lower limit, the sampling is terminated and the whole lot is passed.
- Its main advantage is that it uses smaller sample sizes.
  Take initial sample
  if the number of defectives < lower limit, accept the lot
  if the number of defectives > the upper limit, reject the lot
  if number of defectives between limits, re-sample
  Continue sampling until accept or reject lot based on all sample data.

Sequential Sampling

- It is the extension of multiple sampling plans. In sequential sampling, items are selected from a lot one at a time and after inspection of each sample, a decision is made to accept the lot, reject the lot, or select another item.
- In sequential lot sampling, only a small fraction of the lots submitted are inspected.

Sampling Terms

Certain terms are used for the acceptance sampling plan.

- Acceptance quality level (AQL)
- Lot tolerance percent defective (LTPD) or rejectable quality level (RQL)
- Average outgoing quality (AOQ)
- Producer's risk (α)
- Consumer's risk (β)

Acceptance Quality Level (AQL)

It defines the percentage of defects at which consumers are willing to accept lots as 'good'.

Lot Tolerance Percent Defective (LTPD) or Rejectable Quality Level (RQL)

It defines the upper limit on the percentage of defects that a consumer is willing to accept. The probability of accepting a lot at RQL represents a risk for the consumer. As RQL is an unacceptable quality level, the probability of acceptance on an RQL lot should be below.

Average Outgoing Quality (AOQ)

It represents the average percentage of defectives in the outgoing products after inspection, including all accepted and all rejected lots.

Producer's Risk (α)

It represents the probability that a lot containing the acceptance quality level will be rejected.

Consumer Risk (β)

It represents the probability that a lot containing the defectives exceeding the LTPD will be accepted.

Operating Characteristic (OC Curve) Curve

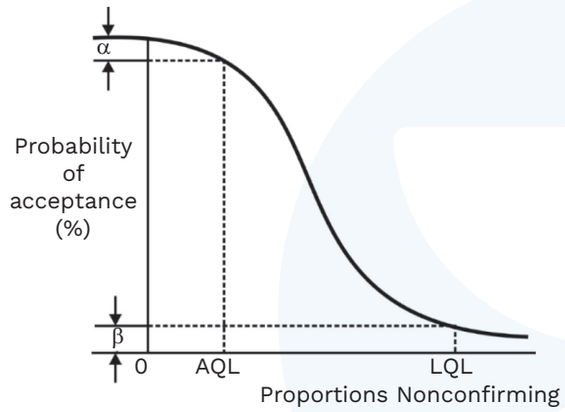


Fig. 7.3

- It is a graph of the % defective (p) in a lot or batch versus the probability that the sampling plan will accept the lot. It shows the probability of log acceptance p_a as a function of lot quality level (p). It is based on the sampling plans.
- OC curve indicates the discriminating power of the plan. It aids in the selection of plans that are effective in reducing risk. It helps to keep the high cost of inspection down.
  Probability of acceptance
  $P_A = P(d \leq C)$
  $$\sum_{d=0}^{C} n_{C_d} p^d (1-p)^{n-d}$$
  p = percent defective in sample size 'n'
  Probability of rejection = $P_R = 1 - P_A$
- OC curve enables the evaluation of the probability of acceptance (P_a) for any true lot quality level on a what-if basis. In this way, one can design sampling plans, that perform the way one wants.

Types of OC Curve

There are two types of OC curves.

- Type A: It gives the probability of acceptance for an individual lot coming from finite production.
- Type B: It gives the probability of acceptance for lots coming from a continuous process or infinite size lot.

Effect on OC Curve

- As the sample size, n increase the OC curve shape will tend to the ideal OC curve more discriminating, and acceptance number C is kept proportional to n.
- Increasing the acceptance number (C) does not significantly change the shape of the operating characteristic curve. When the acceptance number (C) increases keeping the lot size (n) constant, then the producer's risk (α) decreases.

Practice Exercise

1. In acceptance sampling, the producer's risk is the risk of having a:
(A) Good lot accepted
(B) Good lot rejected
(C) Bad lot rejected
(D) Bad lot accepted

2. What is the effect of sample size on the probability of acceptance?
(A) Sample size does not affect the probability of acceptance.
(B) As the number of items in a sample decrease, the higher the possibility of

the number of defectives, thereby the probability of acceptance decreases.

(C) As the number of items in a sample increase, the higher the possibility of many defectives, thereby the probability of acceptance decreases.

(D) None of the above

3. Which of the following statements is/are true?

I. Probability of acceptance decreases if acceptance number increases when sampling size is kept constant

II. Indifferent quality range is the range between lot tolerance percent defective and acceptable quality level

III. Operating characteristics curve is the curve plotted between the probability of acceptance and indifferent quality range

(A) Only I
(B) Only II
(C) Only III
(D) All of the above

4. Maximum data handling analysis load is observed in:

(A) Single sampling plan
(B) Double sampling plan
(C) Sequential sampling plan
(D) All of the above

5. Acceptance sampling of incoming lots is an example of:

(A) End of line quality control or quality assurance
(B) Quality at source
(C) Quality function deployment
(D) Statistical process control

6. In acceptance sampling, when there is a small chance that the lot may be accepted even though the quality is poor, this is referred to as:

(A) Operator's risk
(B) Owner's risk
(C) Consumer's risk
(D) Producer's risk

7. In a double sampling plan, if the number of defects is in between the two cut-off numbers C and C' then:

(A) Accept the lot
(B) Reject the lot
(C) Take another sample
(D) None of the above

8. In a sequential sampling plan:

(A) Maximum 2 samples are taken
(B) Maximum 3 samples are taken
(C) Maximum 4 samples are taken
(D) Samples are accumulated at each stage till the decision of accepting or rejecting a lot is taken

9. Which of the following is not a sampling plan?

(A) Single sampling plan
(B) Double sampling plan
(C) Triple sampling plan
(D) Sequential sampling plan

Answer Key

1. (B) **2.** (C) **3.** (B) **4.** (C) **5.** (D) **6.** (A) **7.** (C) **8.** (D) **9.** (C)

8 Six Sigma

Introduction

- The term 'six sigma' was coined by Bill Smith, father of six sigma, an engineer with Motorola.
- In 1987, Motorola officially launched its six sigma program, and due to the use of six sigma, Motorola is known worldwide as a quality leader. The secret of their success became public knowledge after Motorola won the Malcolm Baldrige National Quality Award in 1988.
- In 1991, Motorola certified its first black belt experts, which indicates the beginning of formalisation of the accredited training of six sigma methods.
- Sigma refers to the distribution or spread of the mean or average of any process or operation. The sigma capability (z-value) is a performance statistic that reflects how well a process is operating. The greater the sigma, the greater the capability.

Six Sigma

- Six sigma is significantly inspired by total quality management (TQM) and zero-defect principles. It argues in its methodology that to attain high-quality manufacturing and commercial processes, continuous efforts to decrease deviations are required.
- The theory behind six sigma is that if you can quantify how many defects (any occurrence or event which does not match customer expectations) you have in a process, you can systematically determine how to eliminate them and reach as near as possible to zero defects. This rise in performance and decrease in process variance leads to a reduction in defects, as well as a substantial boost in earnings and product quality.
- It is a systematic process that employs data (fact-based management) and statistical analysis to monitor and enhance the operational performance, procedures, and systems of a business.
- Processes aiming for six sigma would have no more than 3.4 errors per one million opportunities. The processes can be reviewed to reach the desired level with higher and lower limits. To achieve these constraints, process variation must be minimised. Consequently, the curve will grow more definite

Six Sigma Level	Percentage Accuracy	Defects per Million
6	99.9997	3.4
5	99.98	233
4	99.4	6210
3	93.3	66.807
2	69.1	308537
1	31	698000

Table 8.1

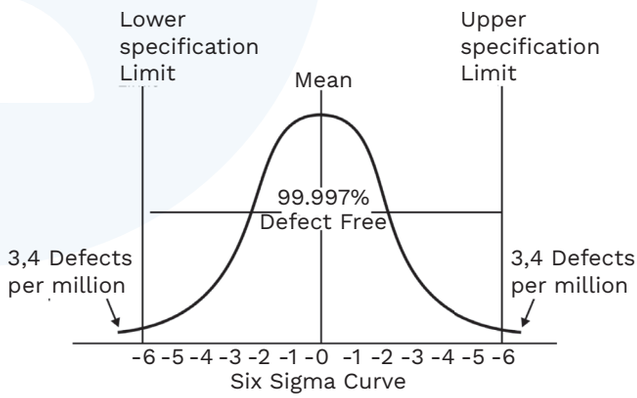


Fig. 8.1

Defects Per Million Opportunities (DPMO)

- To sustain six sigma quality, a process must produce no more than 3.4 defects per million opportunities. Opportunities are the total range of chances per unit to indicate a defect. Each opportunity must be unique, observable, and quantifiable.

- The DPMO may also be considered the process's capabilities. The less the DPMO is involved, the more capable a process is. This can then be converted to a Sigma or standard deviation. The greater the standard deviation, the lower the DPMO, indicating a more reliable process.
- DPMO can be defined as

$$\text{DPMO} = \frac{\text{Number of Defects}}{\text{Number of opportunities for error per unit} \times \text{Number of units}} \times 1000000$$

- The six sigma methodology is normally defined as a set of practices that improve efficiency and eliminate defects. Six sigma project follows two project methodologies—DMAIC and DMADV.

Define, Measure, Analyse, Improve, Control (DMAIC)

- **Define:** The initial step of the procedure is to define. In this step, it is crucial to identify specific objectives for obtaining outcomes that are consistent with both your customer's needs and your business's strategy. Essentially, you are sketching out a plan for achievement.
- **Measure:** To establish whether or not defects have been reduced, a baseline measurement is necessary. In this step, precise measurements and pertinent data must be collected so that future comparison can establish if flaws have been decreased or not.
- **Analyse:** Analysis is essential for identifying causal elements and establishing links. To comprehend how to solve a problem, cause and effect are essential and must be taken into account.
- **Improve:** By improving or enhancing your processes based on measurements and analysis, you can reduce defects and streamline procedures.
- **Control:** It guarantees that deviations are identified and corrected before they can negatively affect a process and cause defects. Controls can take the form of pilot runs to establish if a process is capable, and once data has been gathered, a process can transition into standard production. Nonetheless, ongoing measurement and analysis must be performed to verify that processes remain on track and free of defects below the six sigma limit.

Define, Measure, Analyse, Design, Verify (DMADV)

The demands of the customers concerning quality are not satisfied by the existing process and need to be replaced by a new process rather than simplifying and improving the existing process. At times, an organisation must decide to launch a new product or service to grab a new business opportunity offered by the environment. In all such situations, the last two steps used in the DMAIC, namely, 'improve' and 'control', can be replaced by 'design' and 'verify', so that it becomes DMADV.

- **Define** refers to defining the design goals that are consistent with customer demand and business strategy.
- **Measure** refers to identifying the characteristics that are critical to quality, product capabilities, production process capabilities, and risks.
- **Analyse** refers to developing and designing alternatives.
- **Design** refers to the design of an improved alternative that is best suited to a design perspective.
- **Verify** refers to verifying the design, setting up pilot runs, and implementing the production process.

Different Levels of Six Sigma

- The six sigma belts (green, black, and master black) represent the various levels that can be attained in six sigma. In order to increase the marketability of a company's products and services, it is based on careful and precise planning and consistent, deliberate application.
- The executive leader must be dedicated to this programme and provide the direction necessary to foster teamwork and collaboration among all programme participants. To accomplish this, they must be familiar with the six sigma methodology and choose competent,

dedicated personnel for the champion or sponsor positions.

- **The champion or sponsor** determines what needs to be done and gives the Black Belts both financial resources and dedicated personnel. In addition, this position compares its procedures against those of other businesses to identify areas for improvement.
- **The Master Black Belt** possesses the highest level of knowledge on the subject. They are completely devoted to the methodology and have no other employment obligations. They participate in all aspects of training and coaching for all lower belt ranks. instructors, coaches, and technical leaders are examples.
- A qualified Black Belt demonstrates team leadership, comprehends team dynamics, and assigns roles and tasks to team members. They have a thorough understanding of the DMAIC model following the six sigma principles, a fundamental understanding of lean enterprise concepts, and the ability to swiftly identify non-value-added tasks. Black Belts are generally concerned with project execution, while champions and Master Black Belts identify six sigma projects and functions.
- The DMAIC approach is emphasised in Green Belt. Six sigma Green Belt certification enables a person to act as a trained team member within an organisation's function-specific area. This permits the Green Belt to concentrate on discrete, well-defined six sigma initiatives, taking less time than a Black Belt's full-time commitment to six sigma throughout the entire firm. For example, project and temporary team members.

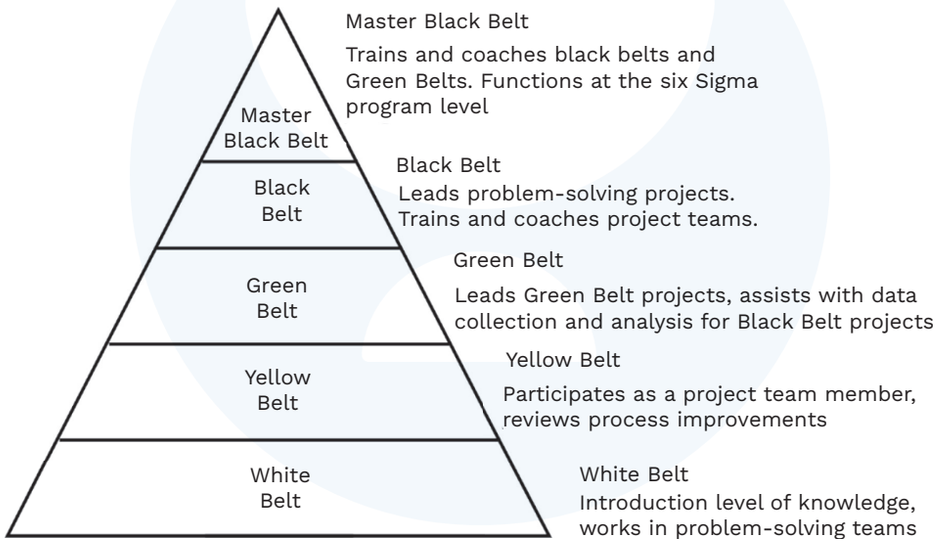


Fig. 8.2

Six Sigma Implementation

The following is a step-by-step approach for implementing six sigma.

- **Step 1:** Every successful performance improvement must start with senior leadership, ensuring that all top-level management is on board and that adequate financial and managerial resources are available. Employee training programmes are implemented after a thorough evaluation of their training requirements.
- **Step 2:** In the second step, define the project's scope and objective based on customer feedback and requirements. Existing projects, surveys, or research might serve as sources of inspiration for six sigma initiatives. Always create goals for the organisation as a whole or for a specific level that requires growth.
- **Step 3:** Perform system analysis to discover faults. Evaluate the present system's flaws and performance. Identify potential

causes of issues. Analyse the impact of potential solutions to the challenges on the organisation.

- **Step 4:** Enhance the system by identifying faster, cheaper, or more efficient alternatives for doing tasks. Implement improvement projects with the aid of management and planning tools.
- **Step 5:** Control the new process, by altering systems and measuring processes, to ensure its success and continuation. Utilise client feedback to identify and resolve upcoming issues.

Practice Exercise

1. Which of the following is the key objective of a six-sigma project?
(A) Developing detailed control charts for critical processes
(B) Developing a matrix to understand the how's and what's of a problem process
(C) Reducing variation in critical processes
(D) Reducing investment costs while improving output quality

2. For a normal distribution, two standard deviations on each side of the mean would include what percentage of the total population?
(A) 95%
(B) 68%
(C) 47%
(D) 34%

3. Six sigma is a business-driven, multi-dimensional structured approach to:
(A) Reducing process variability
(B) Increasing customer satisfaction
(C) Lowering defects
(D) All of the above

4. One of the key roles of a champion or sponsor is:
(A) Hire a team of Master Black Belt, Black Belts, among others
(B) Develop process maps
(C) Perform statistical analysis
(D) Play a pivotal role in that they own the processes of the business and, therefore, must ensure process improvements are captured and sustained

5. They set a very clear scope for all six sigma projects. They are responsible for approving any changes to the scope of the project.
(A) Six sigma deployment leader
(B) Champion or sponsor
(C) Master Black Belt
(D) Black Belt

6. Analyse phase includes:
(A) Identify Vital Project X's and statistically validate them
(B) Communicate and sign off to close project
(C) Generate potential solutions and assess failure mode
(D) All of the above

7. Six sigma gives a precision of:
(A) 95.45%
(B) 99.73%
(C) 99.99%
(D) 68.27%

Answer Key

1. (C) **2.** (A) **3.** (D) **4.** (D) **5.** (B) **6.** (A) **7.** (C)

9 ISO Standards

Introduction

- The International Organisation for Standardisation (ISO), headquartered in Geneva, Switzerland, has developed a set of standards on quality known as ISO standards. Almost all industrialised nations are represented in ISO
- The main aim of ISO Standards is to attain, maintain, and continually improve product quality, as well as to enhance the quality of operations in order to meet the expectations of customers and investors.
- Three parties are involved in the ISO certification such as
  - First party: A firm (the supplier) audits itself against ISO 9000 standards
  - Second party: A customer audits its supplier
  - Third party: A qualified national or international standard or certifying agency serves as an auditor, the basis for getting ISO 9000.

ISO 9000

- ISO 9000 represents a management methodology adopted by companies to deliver products and services that meet customer expectations.
- ISO 9000 defines quality management system standards, based on the premise that certain characteristics of management practices can be standardised, and then a well-designed, and carefully managed quality system provides confidence that the outputs will meet customer expectations and requirements completely.
- ISO 9000 standards' main requirements are system requirements. Management, Resource, Realisation, and Remedial. It directs the organisation to "document what you do and then does as you documented".
- ISO 9000 series are based on the eight quality management principles
- Initially, ISO 9000 standards were divided into six parts: ISO 8402, ISO 9000, ISO 9001, ISO 9002, ISO 9003, and ISO 9004. These standards stand for system standardisation and certification.
- At present, the three main standards of this family are
  - ISO 9000 quality management systems deal with Fundamentals and vocabulary.
  - ISO 9001 quality management systems deal with Requirements.
  - ISO 9004 quality management systems deal with guidelines for performance improvement.
- The ISO 9000 series of standards is a quality management system and an effective management tool. It is a top-down management system that establishes responsibility and empowers the organisation. It respects the organisation's autonomy and it is a tool that permits the organisation to express itself individually.

ISO 9001

- It provides requirements that if met will enable organisations to demonstrate that they can consistently provide a product that meets customer and applicable regulatory requirements. ISO 9001 also states that the standard can be used to assess the organisation's ability to meet customer, regulatory, and organisation's requirements.
- It addresses the needs and expectations of all relevant interested parties and guides the systematic and continual improvement of the organisation's overall performance.

ISO 9004

- It guides for improving the efficiency, effectiveness, and overall performance of an organisation. ISO 9001 and ISO 9004 have been developed as a consistent pair of organisational standards that complement each other.

- ISO 9001 and ISO 9004 share a common structure but can be implemented independently. ISO 9004 is not meant to be a guide to ISO 9001, however, it does contain the requirements of ISO 9001.
- ISO 9001 and ISO 9004 are compatible with one another. Despite having similar construction, they are meant for use together but can also be utilised alone. The current edition of the standards emphasises two important concepts: customer-related processes and continuous improvement.
- The standards have shifted their emphasis from documenting procedures to designing and managing efficient processes. A focus on the function of top management is evaluated alongside a data-driven method for setting measurable objectives and evaluating performance against them. The Shewhart (Deming) plan-do-check-act cycle concepts for quality improvement are incorporated into the standards.
- The new edition of the ISO 9000 standards adheres to a high-level framework with consistent use of key texts and phrases. In addition to an emphasis on a process-oriented approach, risk management, change management, and knowledge management are also covered.

Changes and Current Version of ISO 9000

- As a quality system standard adopted by international organisations for standardisation in 1987, it has been revised many times as in 1999, 2000, 2008, and now presently in 2015.
- Due to these revisions in standards, there have been many changes like changes in titles, changes in content, changes in structure, changes in intent, changes in language, changes in requirements, etc.
- Presently, as per revised in 2015, standards in the ISO 9000 family are:
  - ISO 9000: 2015 - Covers the basic concepts and language
  - ISO 9001:2015 - Sets out requirements of a quality management system.
  - ISO 9004: 2009 - Focuses on how to make a quality management system more effective and efficient.
  - ISO 19011: 2011 - Sets out guidance on the internal and external audits of quality management systems.

ISO 14000

- It is a family of standards that provides a practical tool for companies and organisations to manage their environmental responsibilities.
- ISO 14000 provides standards in three major areas
- Management systems: Systems development and integration of environmental responsibilities into business planning.
- Operations: Consumption of natural resources and energy
- Environmental systems: Measuring, assessing, and managing emissions, effluents, and other waste.
- ISO Technical Committee 207 is responsible for the ISO 14000 Series of standards and ISO 14000 is based on the British standard BS 7750 which was instituted in 1992.
- The actual environment standards of ISO 14000 deal with how a company manages the environment inside its facilities and the immediate outside of the company.
- ISO 14000 standards do not mandate a particular level of pollution or performance, but they force awareness of the processes and procedures that can affect the environment.
- Important standards of ISO 14000 series are as follows.
  - ISO 14001 - specification of environmental management systems
  - ISO 14004 - guideline standard
  - ISO 14010 to ISO 14015 - environmental auditing and related activities
  - ISO 14020 to ISO 14024 - environmental labelling ISO 14031 to ISO 14032 - Environmental performance Evaluation
  - ISO 14040 to ISO 14043 - life cycle assessment
  - ISO 14050 - terms and definitions
- All ISO standards are reviewed every five years (to establish if a revision is required) to keep them current and relevant for the marketplace.
- The revision is conducted by an ISO technical committee, which is comprised of experts

nominated by their National Standard Bodies and liaison organisations.

- ISO 14001:2015 is designed to respond to the latest trends and ensure it is compatible with other management system standards.
- Ford Motor Company is one of the leading corporations in the use of environmental management systems. Not only are all of its plants across the world ISO 14001-certified, but they have also made great strides in air pollution, energy usage, recycling, waste disposal, and water treatment. Significant recycling initiatives have prevented solid waste from entering landfills. Ford has supported the spread of environmental awareness by offering monetary incentives to suppliers that contribute to its waste reduction programme.

ISO 14001: 2015

- Key changes in ISO 14001:2015 are mentioned below.
  - a greater emphasis on environmental management in the organisation's strategic planning processes.
  - Addition of proactive activities to safeguard the environment from damage and degradation, including sustainable resource utilisation and climate change mitigation
  - Considering environmental factors in terms of the lifecycle
- ISO 14001:2015 is the most popular standard of the ISO 14000 family, which also includes standards, such as:
  - ISO 14004 - general guidelines on principles, systems, and support
  - ISO 14006 - guidelines for incorporating eco-design techniques
  - ISO 14015 - environmental assessment of sites and organisations
  - ISO 14020 - environmental labels and declarations
  - ISO 14031 - environmental performance evaluation
  - ISO 14040 - life cycle assessment
  - ISO 14050 - vocabulary
  - ISO 14063 environmental communication
  - ISO 14064 - greenhouse gases
  - ISO 19011 - guidelines for auditing management systems

OHSAS 18000

Occupational, health, and safety management standards (OHSAS) are international standards that provide a framework to identify, control, and decrease the risks associated with health and safety within a company. Implementing this standard gives a clear signal to investors that management view employees' health and safety as a principal priority within the company.

OHSAS 18001

- It is worldwide known as the occupational, health, and safety management system series standard. It is based on the similar ISO 9001 and ISO 14001 Plan-Do-Check-Act Structure. To achieve certification of OHSAS 18001 an organisation must have a healthy and safety management system compliant with the requirement of the OHSAS 18001 standard.
- It provides organisations with a framework for health and safety management helping them to identify, minimise and control health and safety risks, conform to health and safety legislation, and Protect the wellbeing of employees.

Emission Norms

Auto emission norms are emission standards issued by a country's government to regulate the air pollutants emitted by internal combustion engine equipment including automobiles.

Euro Norms

- In European nations, the regulatory standards are referred to as Euro Norms, denoted by the initials EU followed by the numerals I, II, III, and IV in ascending order of strictness. Even though vehicle standards vary from country to country due to differences in traffic density and regulation, road infrastructure, fuel quality, and maintenance standards, India uses Euro standards with one modification: a lower maximum speed of 90 km/h in the final part of the cycle as opposed to the EU-mandated 120 km/h. It is known as the modified European cycle.

- Euro norms refer to the permissible emission levels from both petrol and diesel vehicles. In India, the government has adopted the Euro norms for available fuel quality and the method of testing. Euro-1 norms in India are known as INDIA 2000 since it was implemented on 1/4/2000. The norms equivalent to Euro-2 are called 2005 norms.

Bharat Stage Norms

- Bharat Stage Emission Standards are the standards established by the Indian government that regulate the amount of air pollutants that can be emitted from internal combustion (IC) engines and vehicles. These cannot be sold on the open market if they release more pollutants than the authorised level. The Central Pollution Control Board (CPCB) of the Ministry of Environment, Forests, and Climate Change has enacted these regulations.
- In 1991, India adopted pollution standards for petrol vehicles, and in 1992, diesel vehicles. Since the year 2000, four-wheeled cars in India have adhered to Bharat Stage Emission Standards, also known as Euro Standards.
- Bharat stage III norms have been enforced across India since October 2010. The recommendation is based on the Mashelkar committee report for the auto-fuel policy launched in the country for improvement of air quality, In a few cities, Bharat stage IV norms placed since April 2010. Bharat stage IV is proposed to be enforced throughout India by April 2017. It has already been put into use in 13 major cities.
- Bharat stage VI norms will be implemented from 1 April 2020. specifically, gasses like carbon dioxide (CO_2), carbon monoxide (CO), hydrocarbons, and oxides of nitrogen, which are the by-products of a running IC engine. It is said that the emissions of nitrogen oxide compounds will be brought down by 68% in diesel cars and 25% in petrol cars.

Bharat Stage Norms vs Euro Norms

- The European rules for new heavy-duty diesel engines are typically referred to as Euro norms, whilst BS-II and BS-III are its evolved variants. Even if the emission regulations are the same, the variances stem mostly from environmental and geographical considerations.
- Euro-III is tested in European countries in sub-zero temperatures. In India, where the yearly average temperature ranges from 24 to 28 degrees Celsius, the test is no longer administered.
- The maximum speed at which a vehicle is evaluated is an additional important distinction. BS-II specifies a maximum speed of 90 km/h, but Euro-II specifies a maximum speed of 120 km/h, with emission limitations remaining unchanged.
- In addition to limits, the test technique also includes certain finer details. In Europe, for example, mass emission test measurements conducted in g/km on a chassis dynamometer require the addition of 100 kg to the unloaded vehicle weight. Due to the road conditions in India, BS-III regulations require an additional 150 kg of weight to reach the desired inertia weight.
- The remaining components of the driving cycle, such as the idling phase, acceleration and deceleration rates, durations, and the cruising period, are identical to European practice.

Practice Exercise

1. ISO:9000 quality system contains:

I. Legal provision
II. Measurement
III. Document record
IV. Standardisation

(A) I, II, and III
(B) I, III, and IV
(C) I, II, and IV
(D) II, III, and IV

2. Match List I with List II and select the correct answer using the codes given below the lists.

List I	List II
A. ISO:9001	I. Bridge
B. ISO:9002	II. Heat exchanger
C. ISO:9003	III. Petroleum products

Codes

	A	B	C
(A)	I	II	III
(B)	III	II	I
(C)	II	I	III
(D)	II	III	I

3. In an ISO 14001. Conforming EMS, which of the following documents should be controlled?
(A) Regulatory permit
(B) Emergency response plan
(C) Both (A) and (B)
(D) None of the above

4. Roll-out dates for BSIV and BS V are:
(A) 2019, 2021
(B) 2017, 2020
(C) 2021, 2024
(D) 2020, 2023

5. It is a requirement of ISO 14001 that an organisation implementing an Environmental Management System should set Environmental targets to achieve environmental objectives within a specified timeframe. These targets should be:
(A) Implemented on a trial basis before taken into force
(B) Related to financial effects
(C) Generic in nature
(D) Specific and measurable

6. Focusing on reducing carbon production to reduce carbon dioxide emissions is argued to be the wrong variable to focus on because
(A) It is impossible to achieve.
(B) The focus should be on carbon consumption.
(C) It has nothing to do with carbon dioxide emissions.
(D) The focus should be on carbon elimination.

7. ISO 14001 is:
(A) A particularly harmful global warming gas
(B) A European Union regulation on pollution abatement
(C) A water quality regulation set out under the UK 1996 Environment Act
(D) An international standard setting out best practices for carrying out audits by companies of their environmental management systems and policies

8. How often should the organisation's environmental management system be reviewed by the top management?
(A) Every six months, as long as every member of the board is present
(B) Once a year
(C) At planned intervals that the top management will decide
(D) After the completion of each internal audit

9. Which of the following is the explanation of an environmental objective?
(A) Management target for quality
(B) Overall financial and business goals
(C) Overall environmental goal, consistent with the environmental policy, that an organisation sets itself to achieve
(D) Surroundings in which an organisation operates

10. An environmental impact is:
(A) Those activities, products, or services which can pollute the environment
(B) Poor environmental management that results in pollution
(C) Any change to the environment, whether adverse or beneficial
(D) All of the above

11. ISO 14001 requires an organisation to:
(A) Recycle waste paper
(B) Conduct environmental assessments of suppliers' management systems
(C) Measure the amount of electrical power used
(D) None of the above

12. An organisation's EMS resources include:

I. Infrastructure
II. Human resources
III. Technology

(A) I only
(B) II only
(C) III only
(D) I, II, and III

13. ISO 14001 requires an organisation to:

(A) Set improvement targets for every department
(B) Define a quality policy within the defined scope of its environmental management system
(C) Define the scope of its environmental management system
(D) None of the above

14. Which of the following must be included in an environmental policy to comply with ISO 14001?

(A) Commitment to continual improvement and prevention of pollution
(B) Statement that determines all the applicable legislation
(C) Both (A) and (B)
(D) None of the above

Answer Key

1. (D) **2.** (C) **3.** (C) **4.** (A) **5.** (C) **6.** (B) **7.** (D) **8.** (C) **9.** (C) **10.** (C)
11. (D) **12.** (D) **13.** (C) **14.** (A)